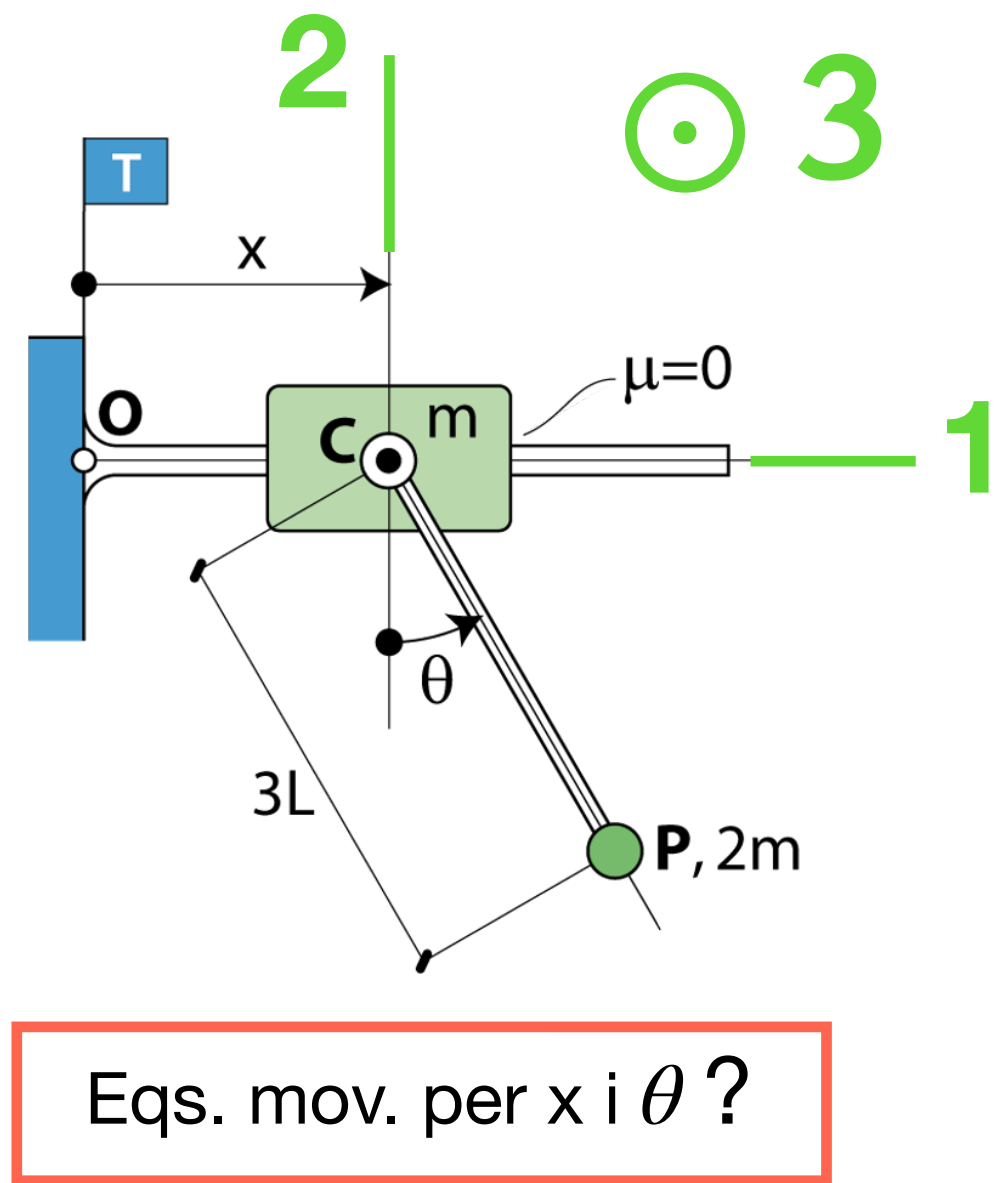
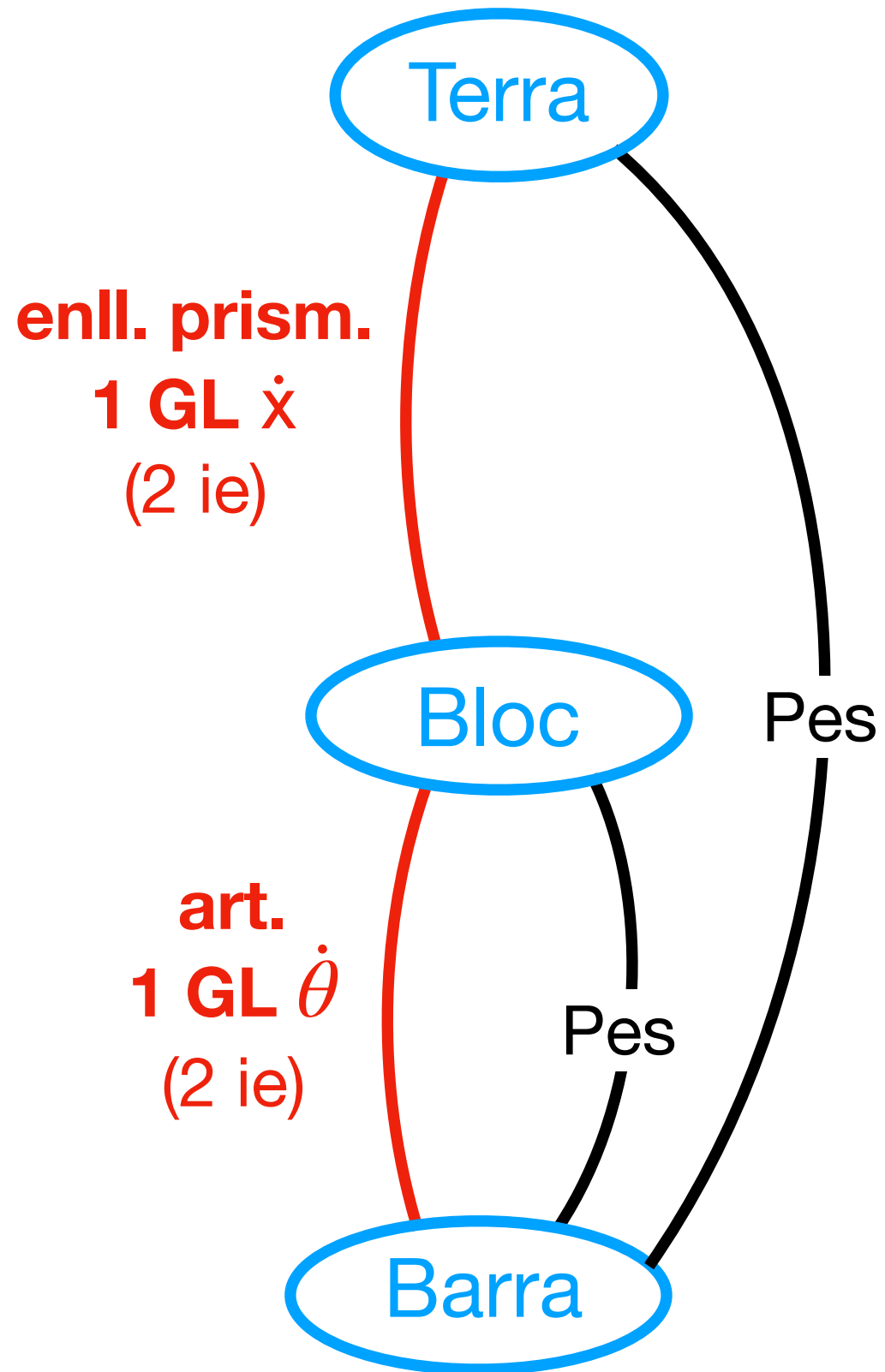


12P

Teoremes vectorials III

Exemples 3D

Recordem: Pèndol d'Euler



DGIE

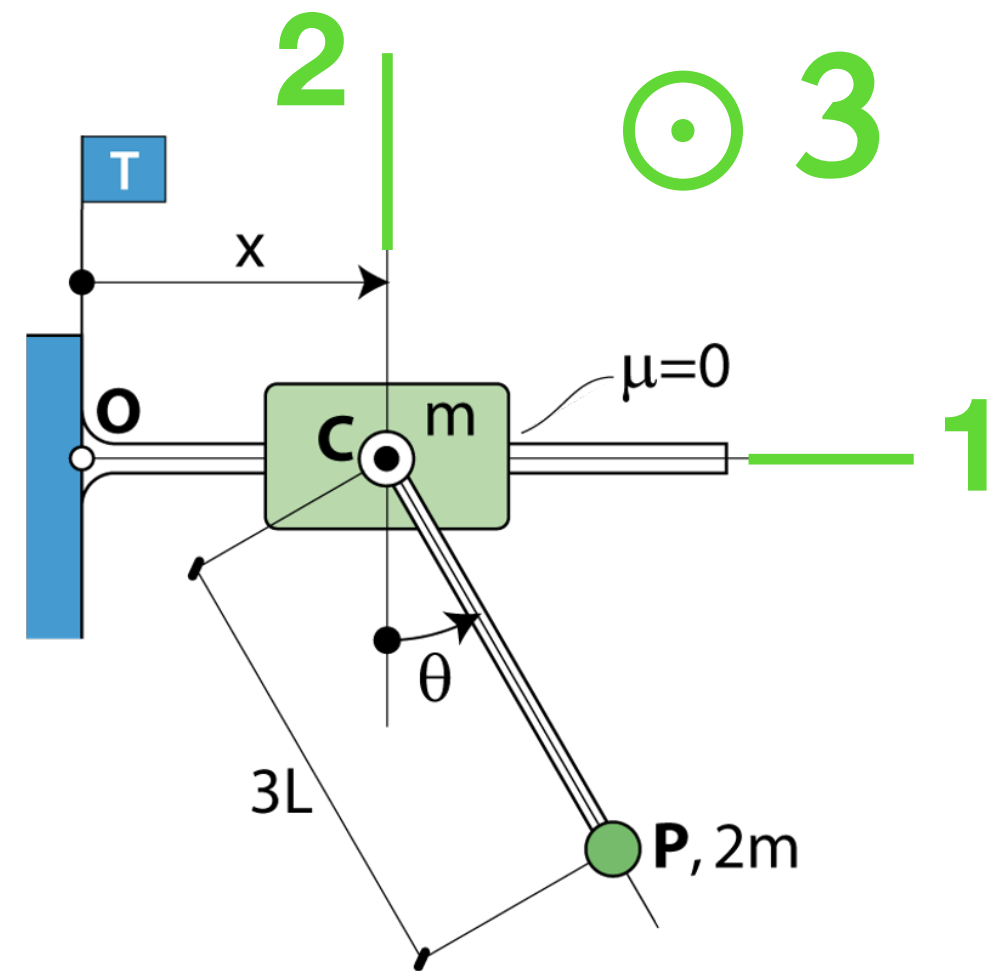
Terra

enll. prism.
1 GL $\dot{x}$
(2 ie)

Bloc

art.
1 GL $\dot{\theta}$
(2 ie)

Barra



TQM $]$ $_1$ condueix a
eq. mov. per a x

TMC(**C**) $]$ $_3$ condueix a
eq. mov. per a θ

Terra

$\dot{q}_n$

S_n

⋮

$\dot{q}_3$

S_3

$\dot{q}_2$

S_2

$\dot{q}_1$

S_1

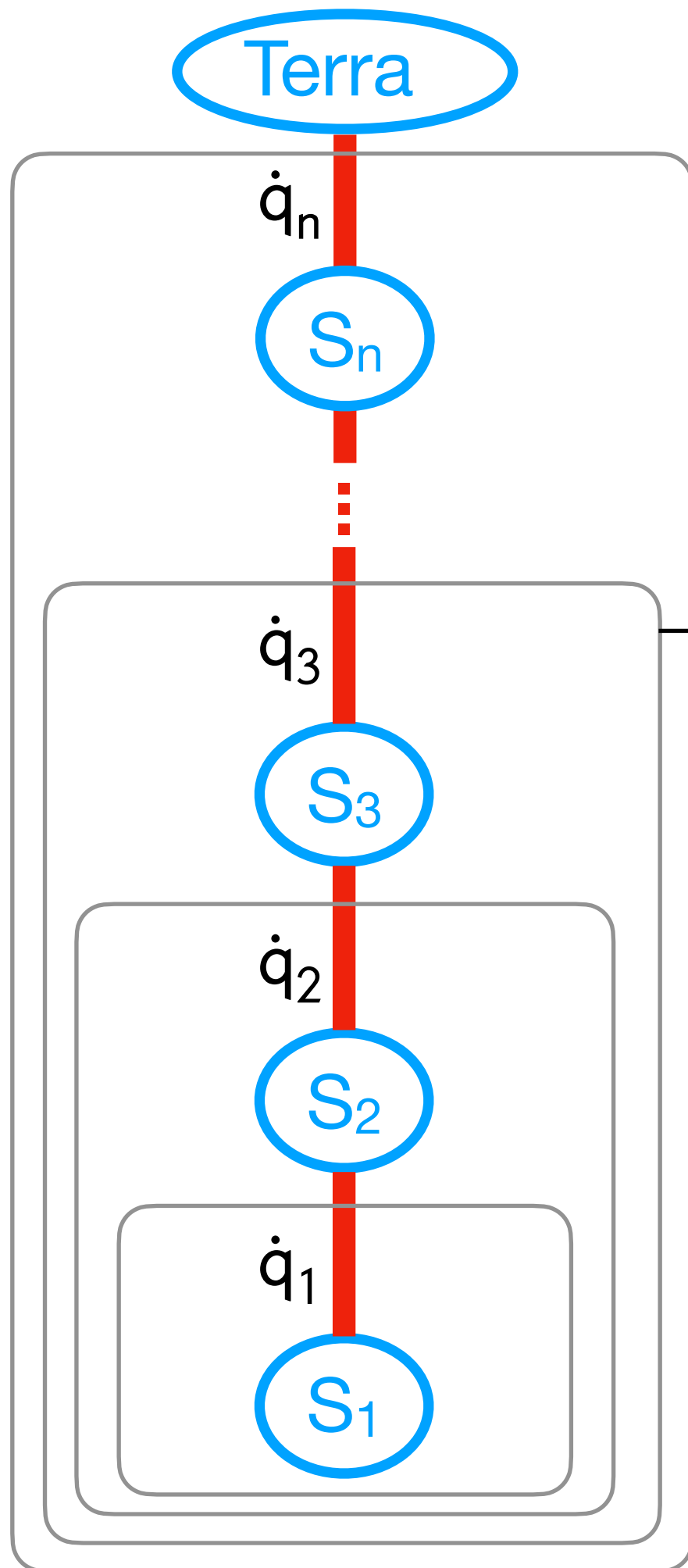
Eq. mov. per q_n

⋮

Eq. mov. per q_3

Eq. mov. per q_2

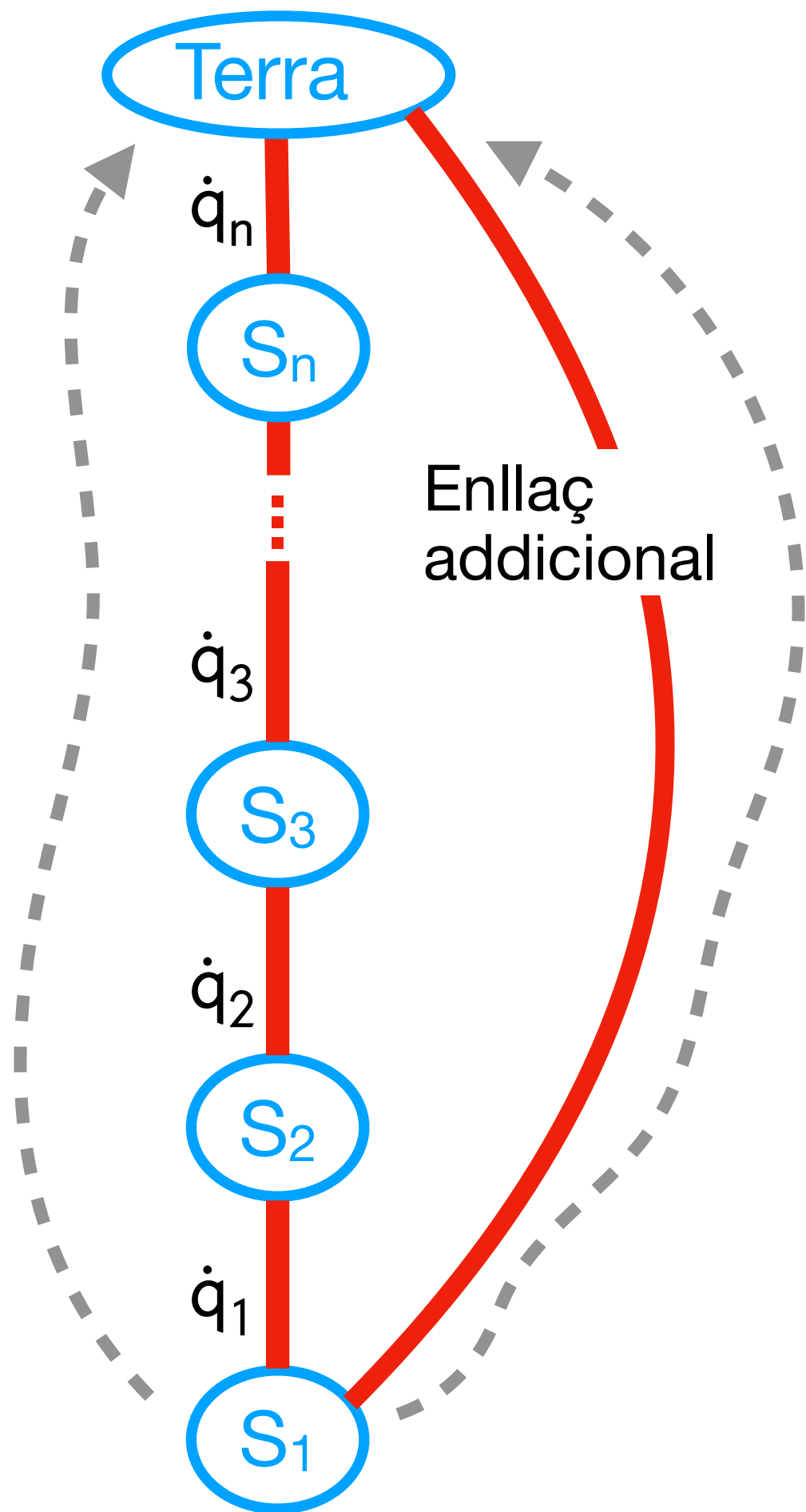
Eq. mov. per q_1



Si un motor garanteix $\dot{q}_3 = ct$, el

$$\text{Sistema} = S_1 + S_2 + S_3$$

no dóna una equació del moviment, sinó la força o parell motor que assegura $\dot{q}_3 = ct$

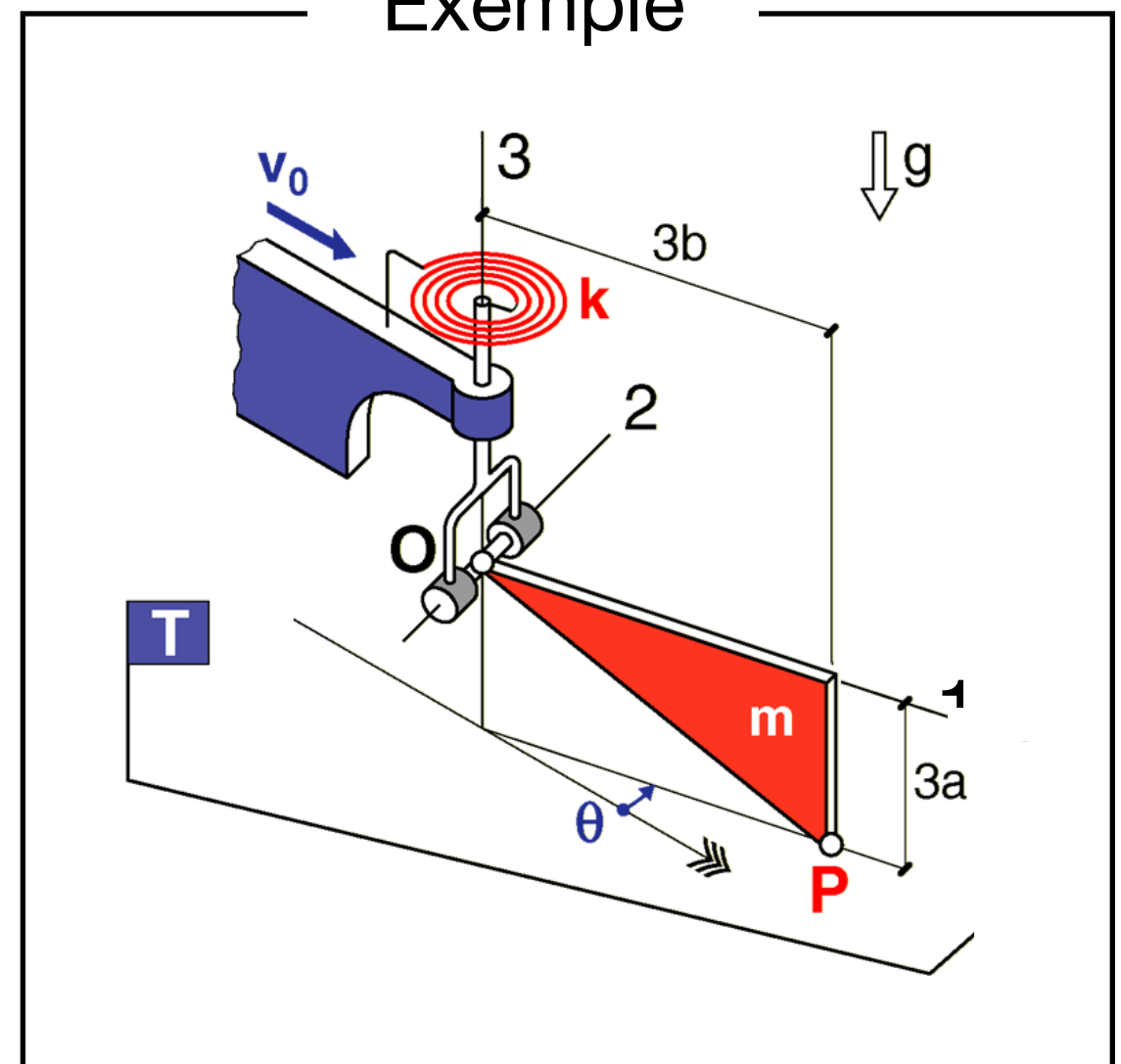


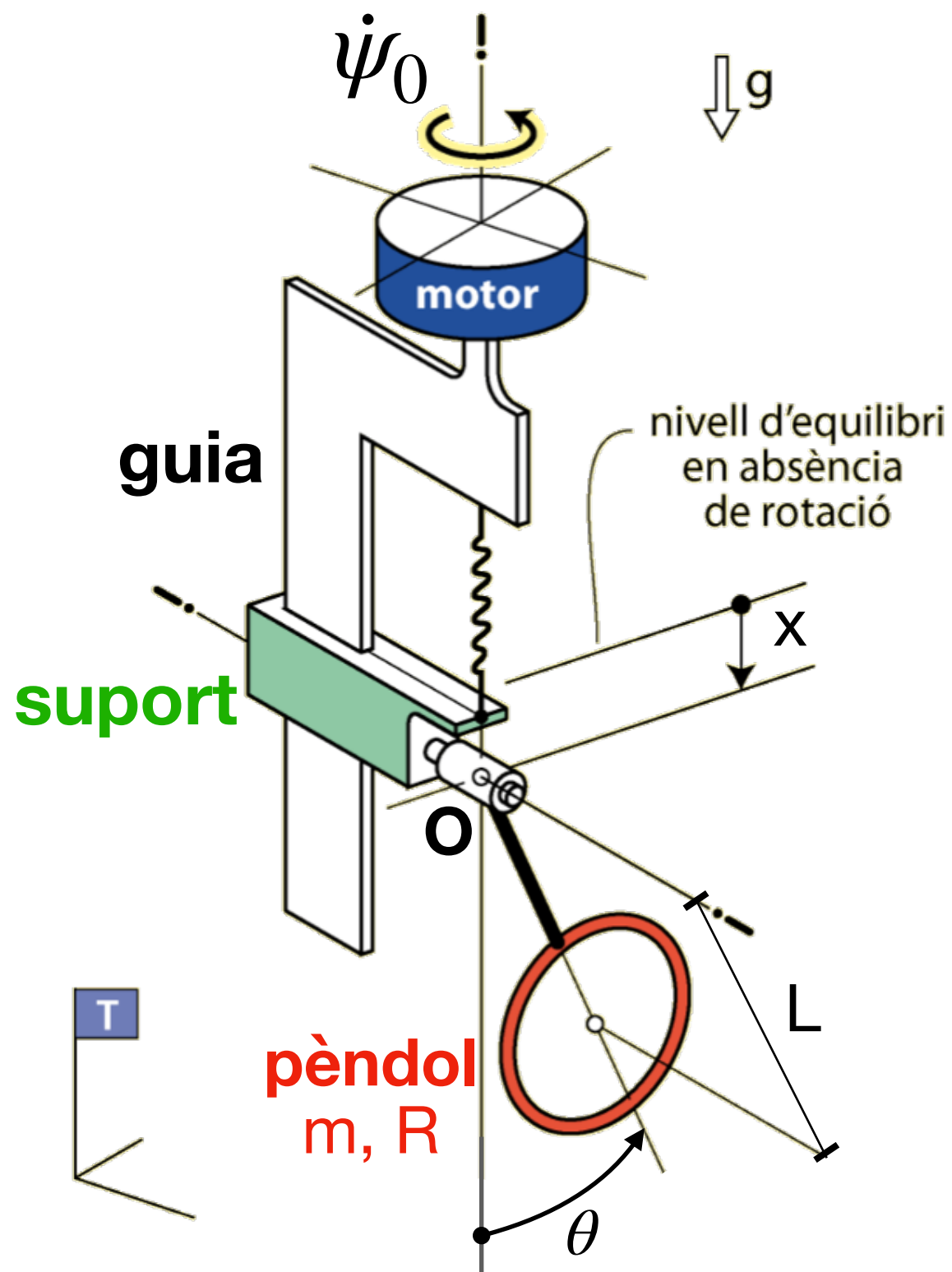
Camins múltiples entre S_1 i Terra



No podem aplicar els coneixements previs

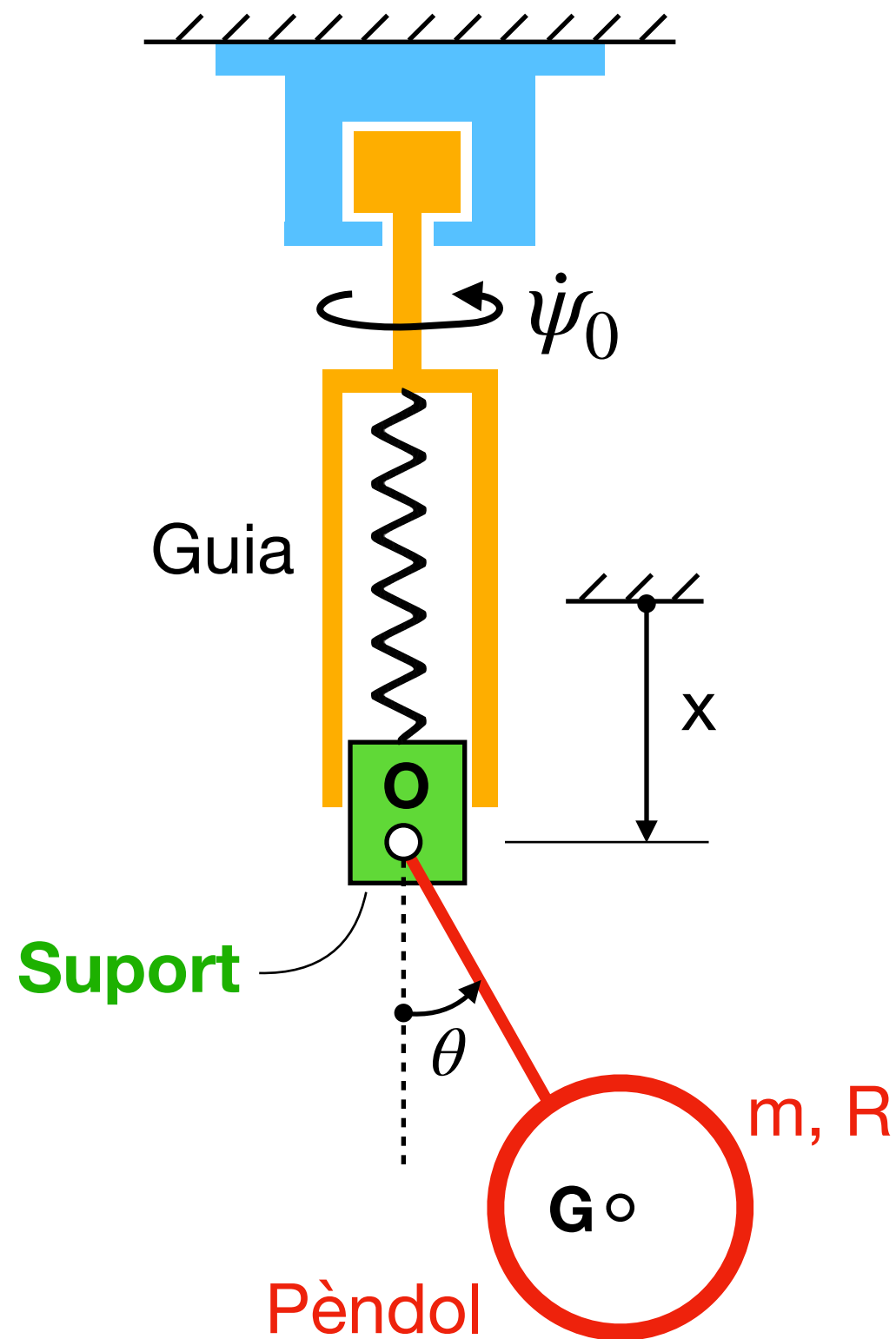
Exemple





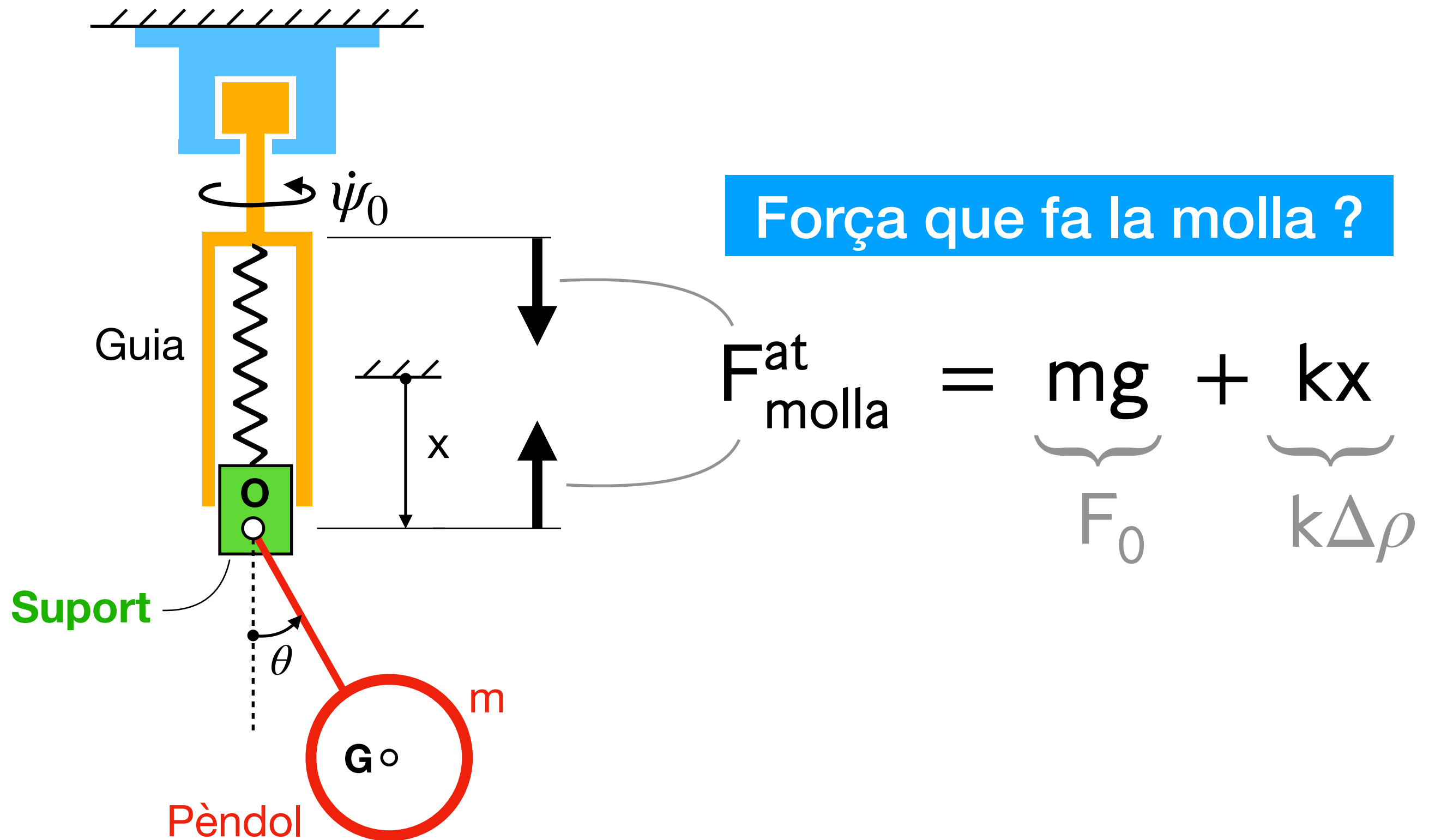
- DGI
- Eq. mov. per a x
- Eq. del mov per a θ
- Parell motor Γ per mantenir $\dot{\psi}_0 = \text{ct}$

Amb motor aturat: $x = 0$, $\theta = 0$ és config. d'equilibri



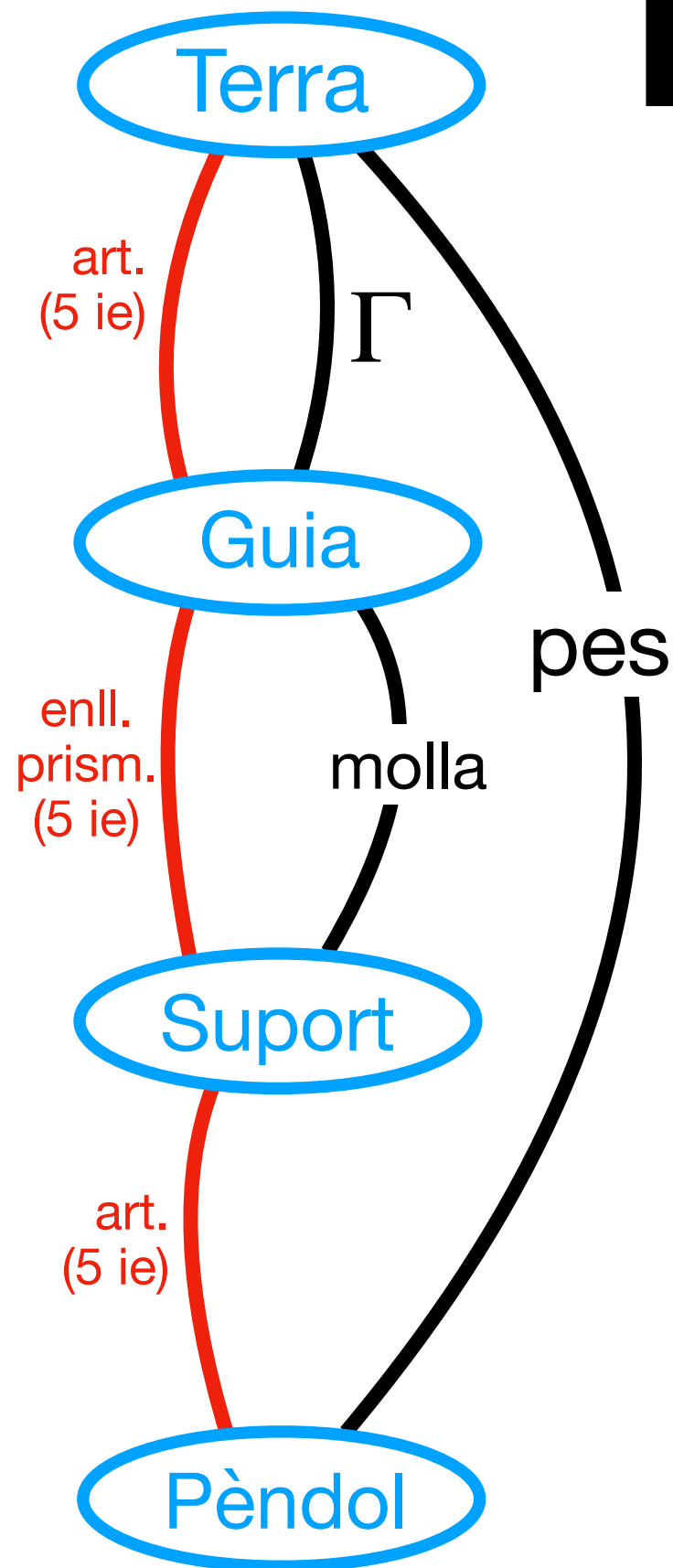
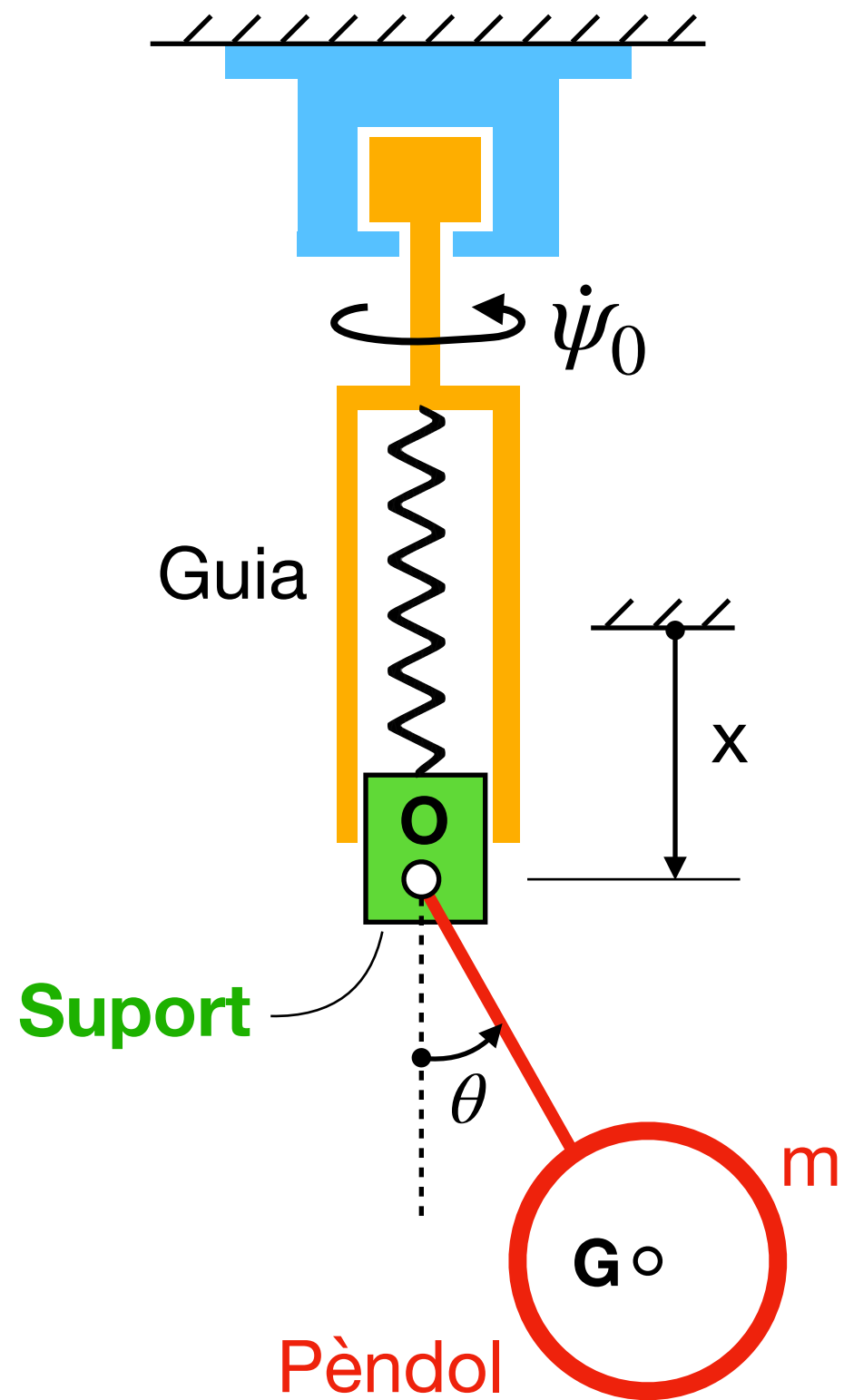
- DGI
- Eq. mov. per a x
- Eq. del mov per a θ
- Parell motor Γ per mantenir $\dot{\psi}_0 = ct$

Amb motor aturat: $x = 0$, $\theta = 0$ és config. d'equilibri

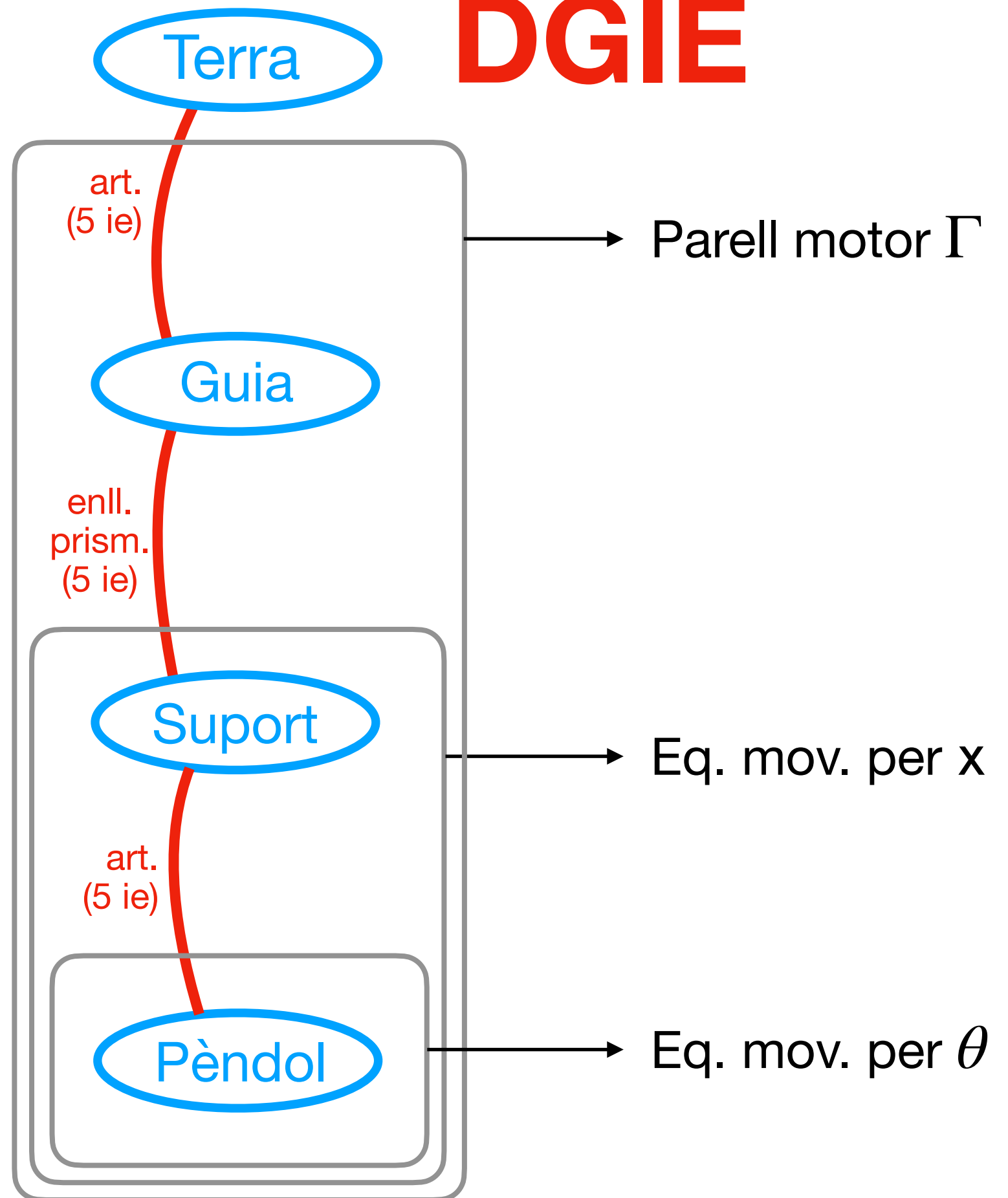
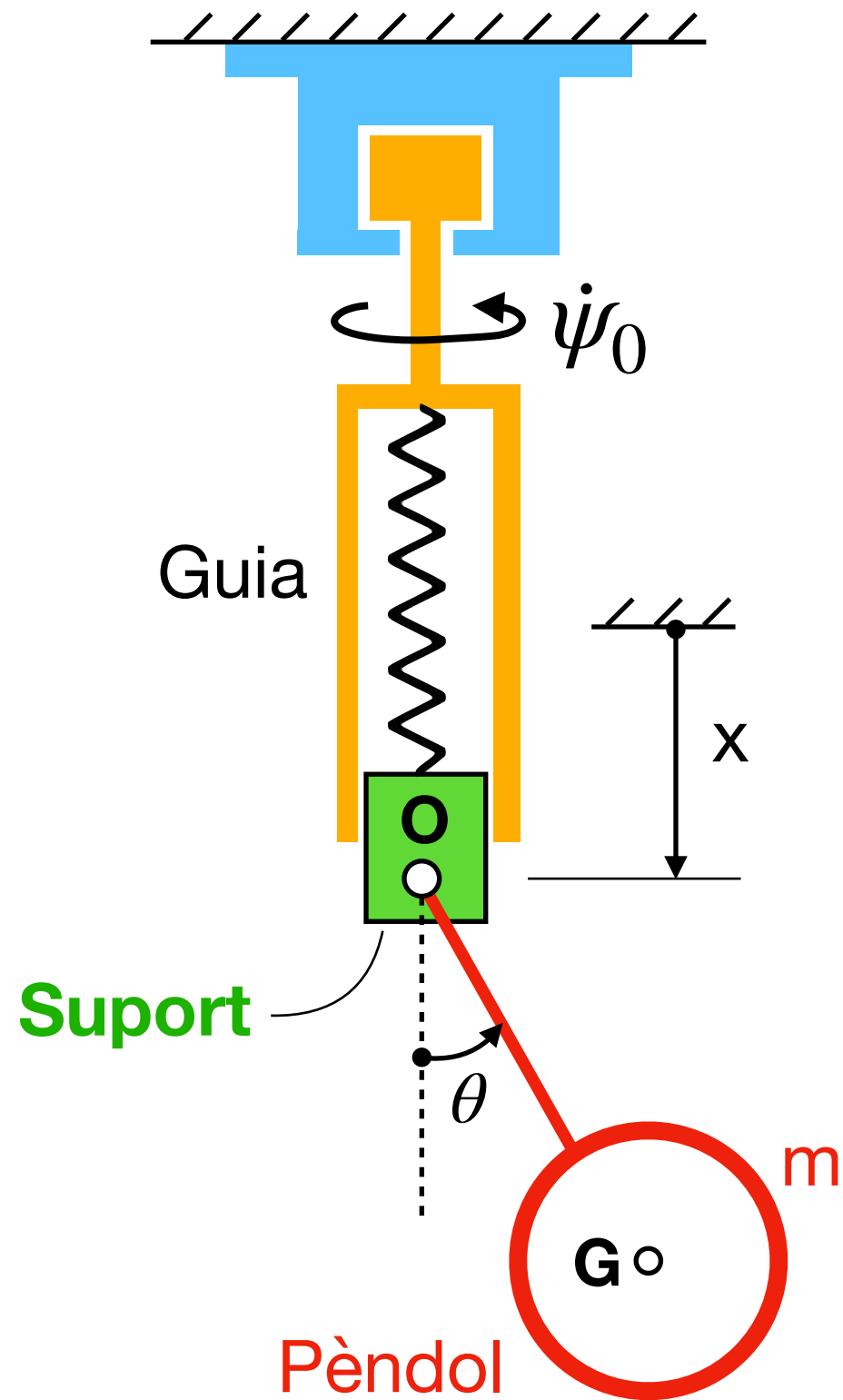


Amb motor aturat: $x = 0$, $\theta = 0$ és config. d'equilibri

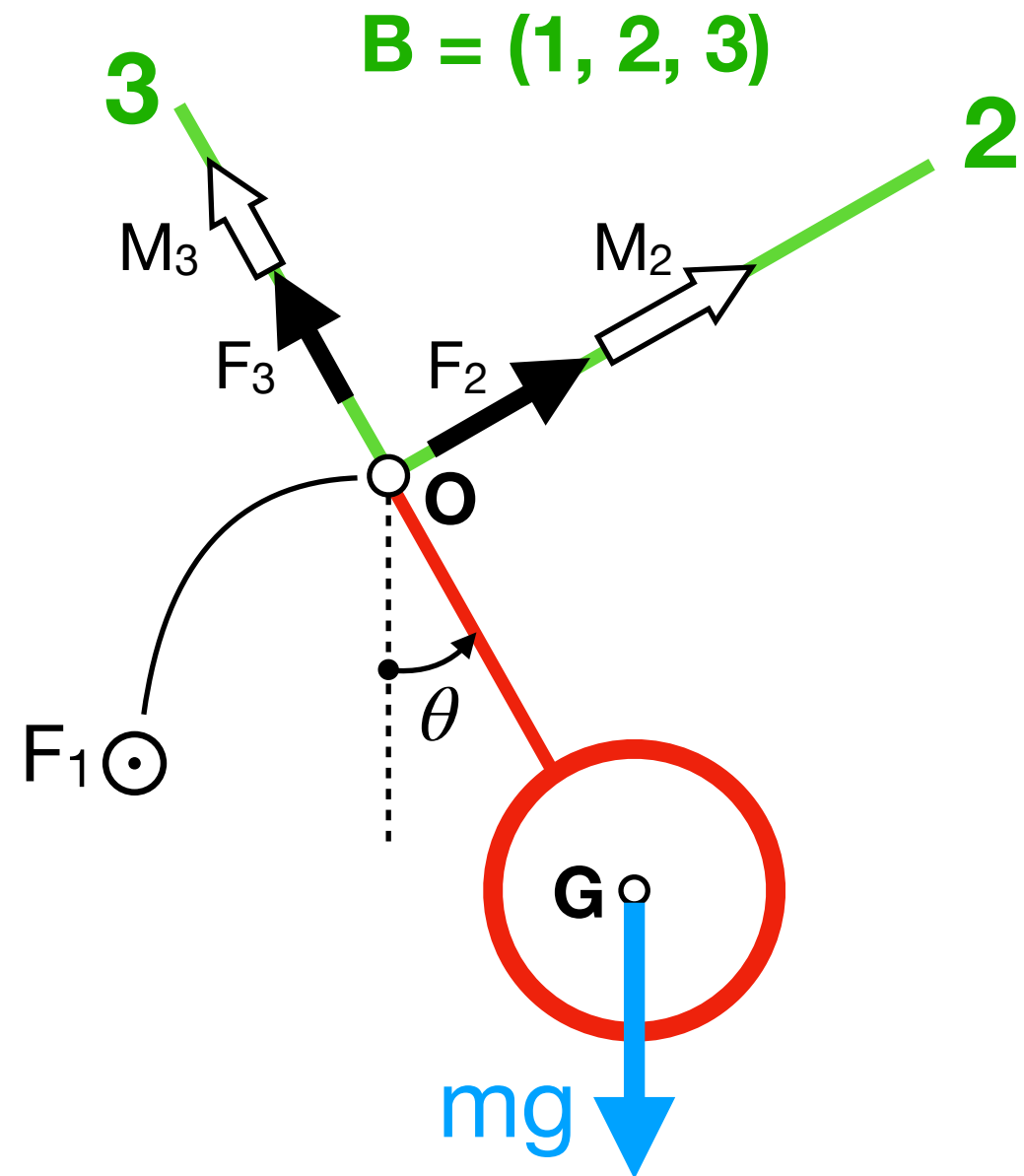
DGI



DGIE



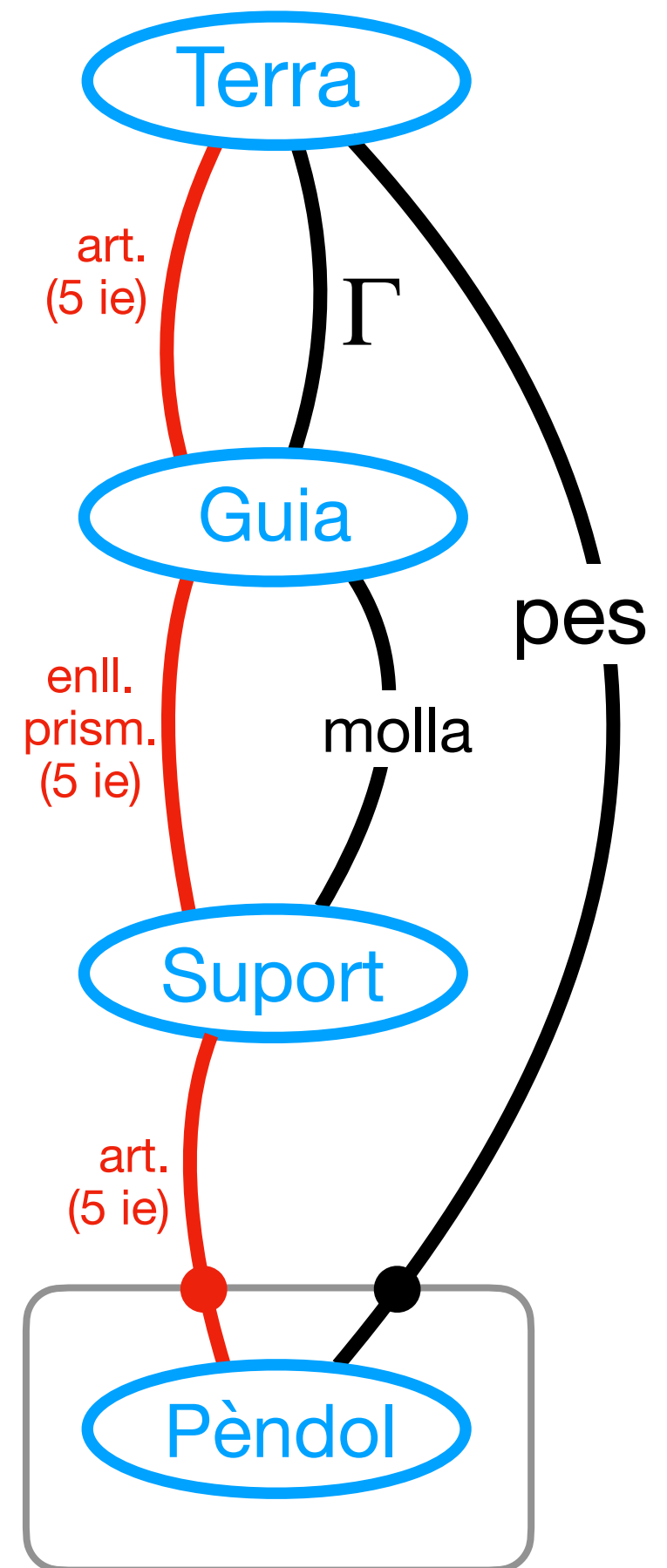
Sist = Pèndol



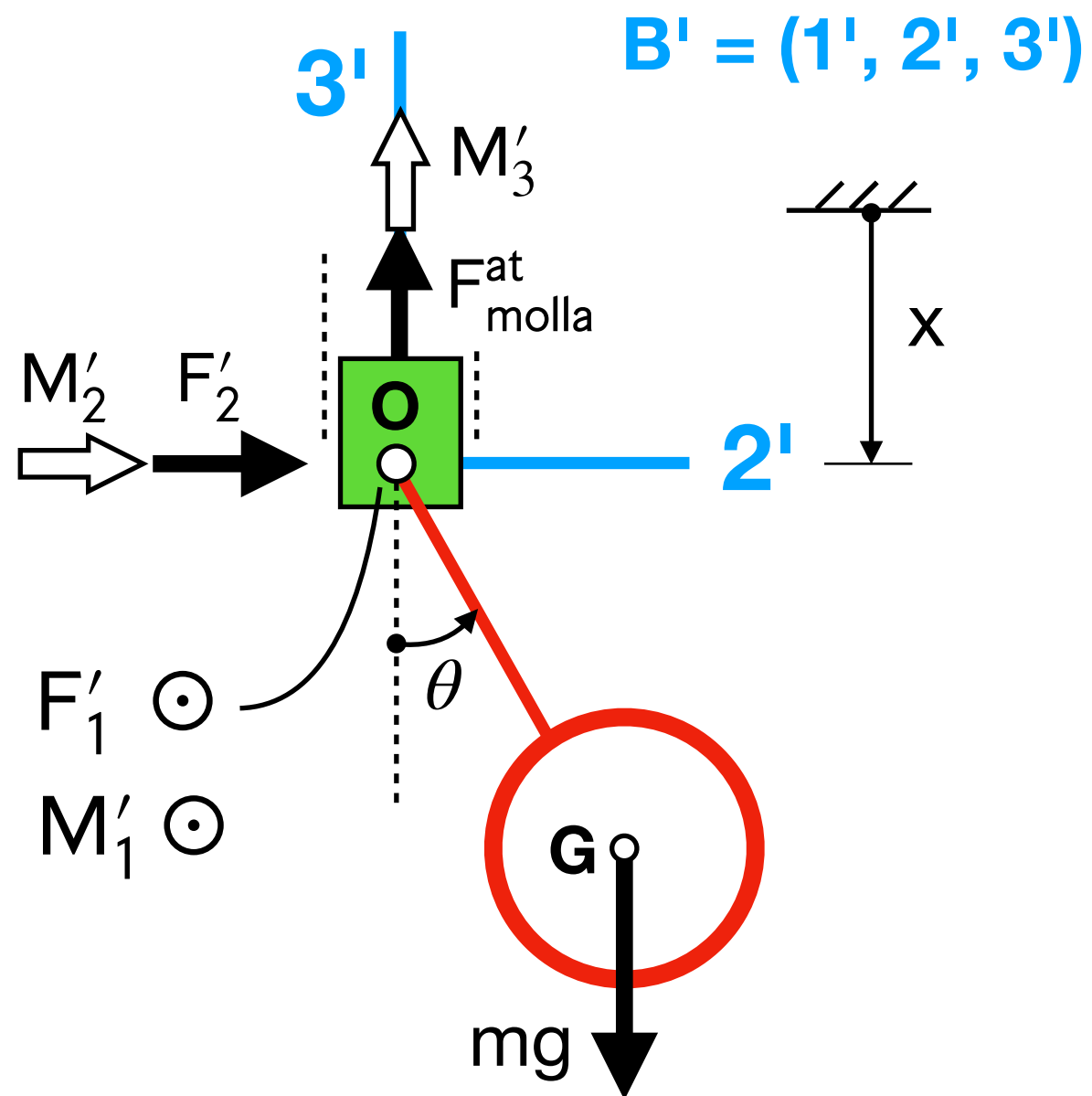
$$\left\{ \bar{F}_{\text{Sup} \rightarrow \text{Pendol}} \right\}_B = \begin{Bmatrix} F_1 \\ F_2 \\ F_3 \end{Bmatrix}$$

$$\left\{ \bar{M}_{\text{Sup} \rightarrow \text{Pendol}}(\mathbf{O}) \right\}_B = \begin{Bmatrix} 0 \\ M_2 \\ M_3 \end{Bmatrix} \leftarrow \text{TMC}(\mathbf{O}) \Big|_1$$

lliure de ie



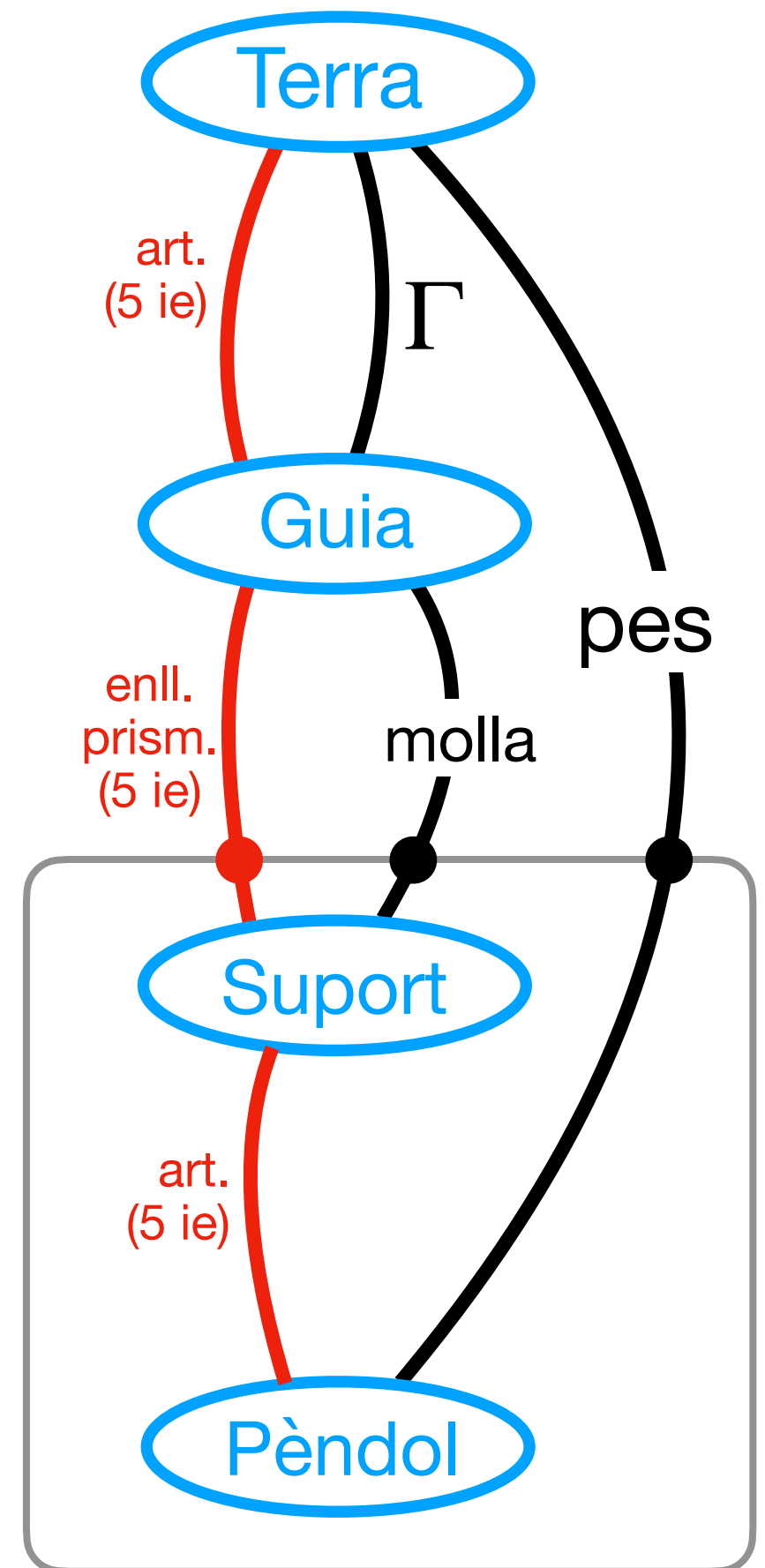
Sist = Pèndol + Suport



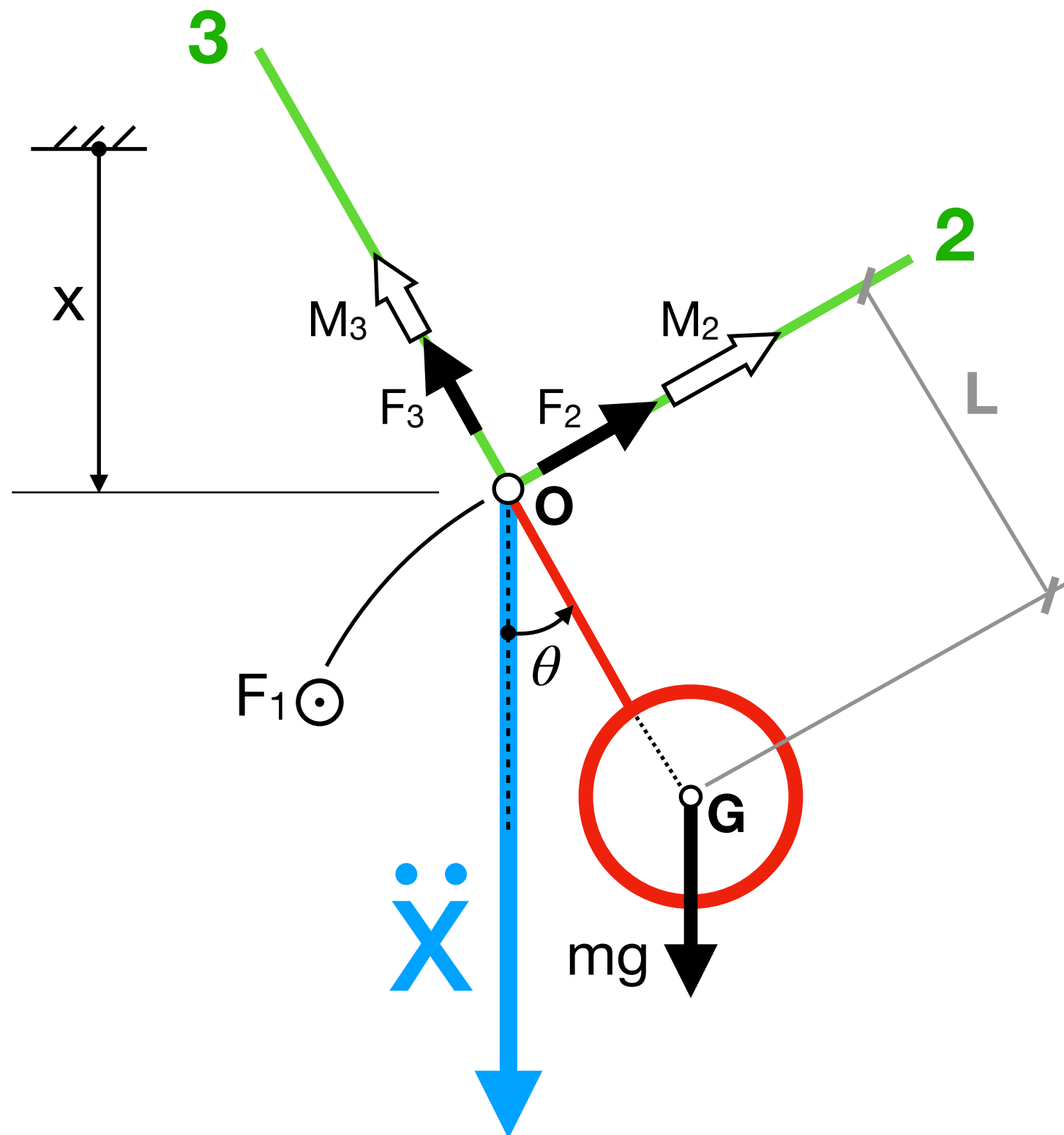
$$\left\{ \bar{F}_{\text{Guia} \rightarrow \text{Sup}} \right\}_{B'} = \begin{Bmatrix} F'_1 \\ F'_2 \\ 0 \end{Bmatrix}$$

$$\left\{ \bar{M}_{\text{Guia} \rightarrow \text{Sup}}(\mathbf{O}) \right\}_{B'} = \begin{Bmatrix} M'_1 \\ M'_2 \\ M'_3 \end{Bmatrix}$$

TQM]_{3'}
lliure de ie

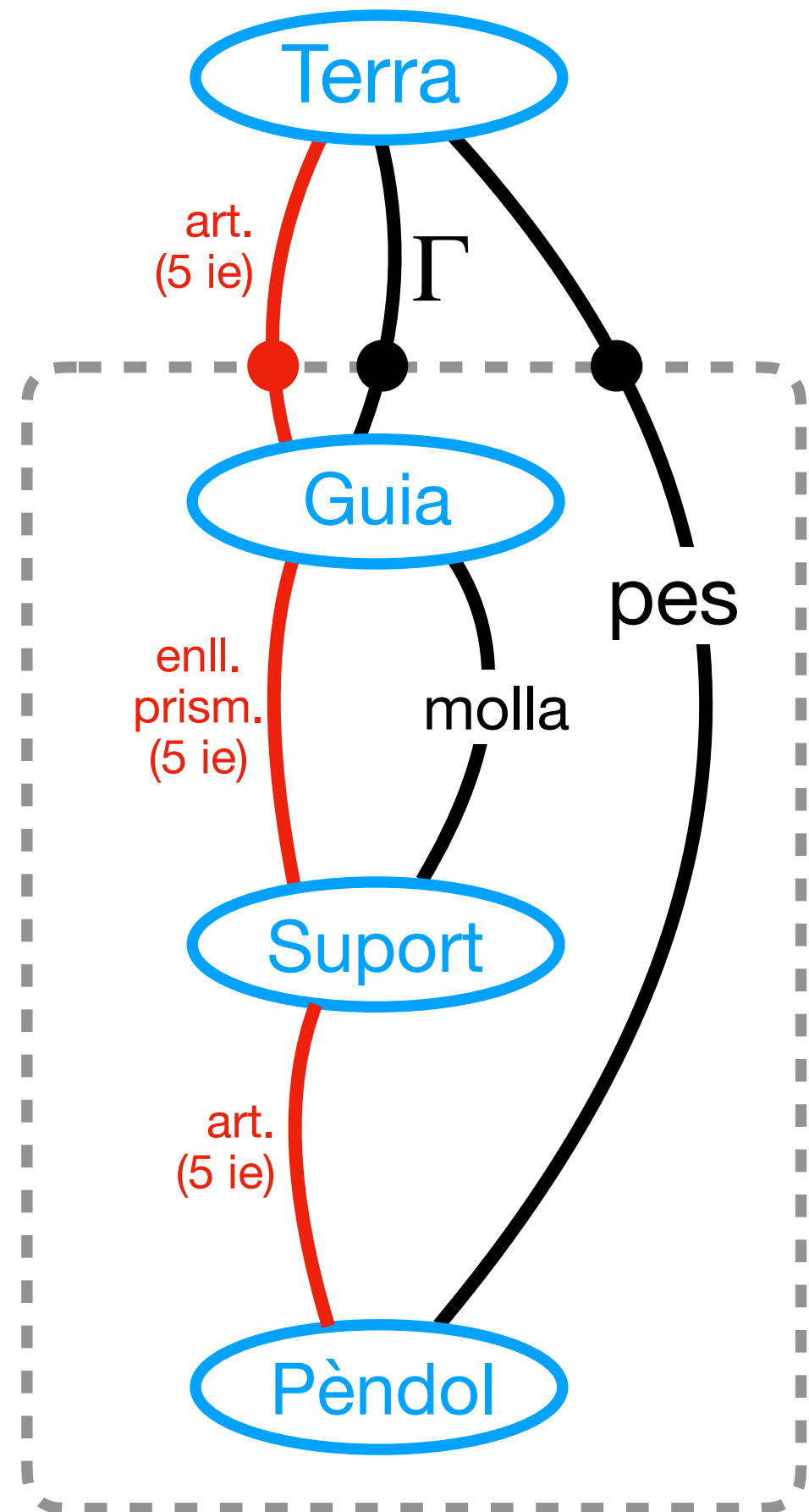
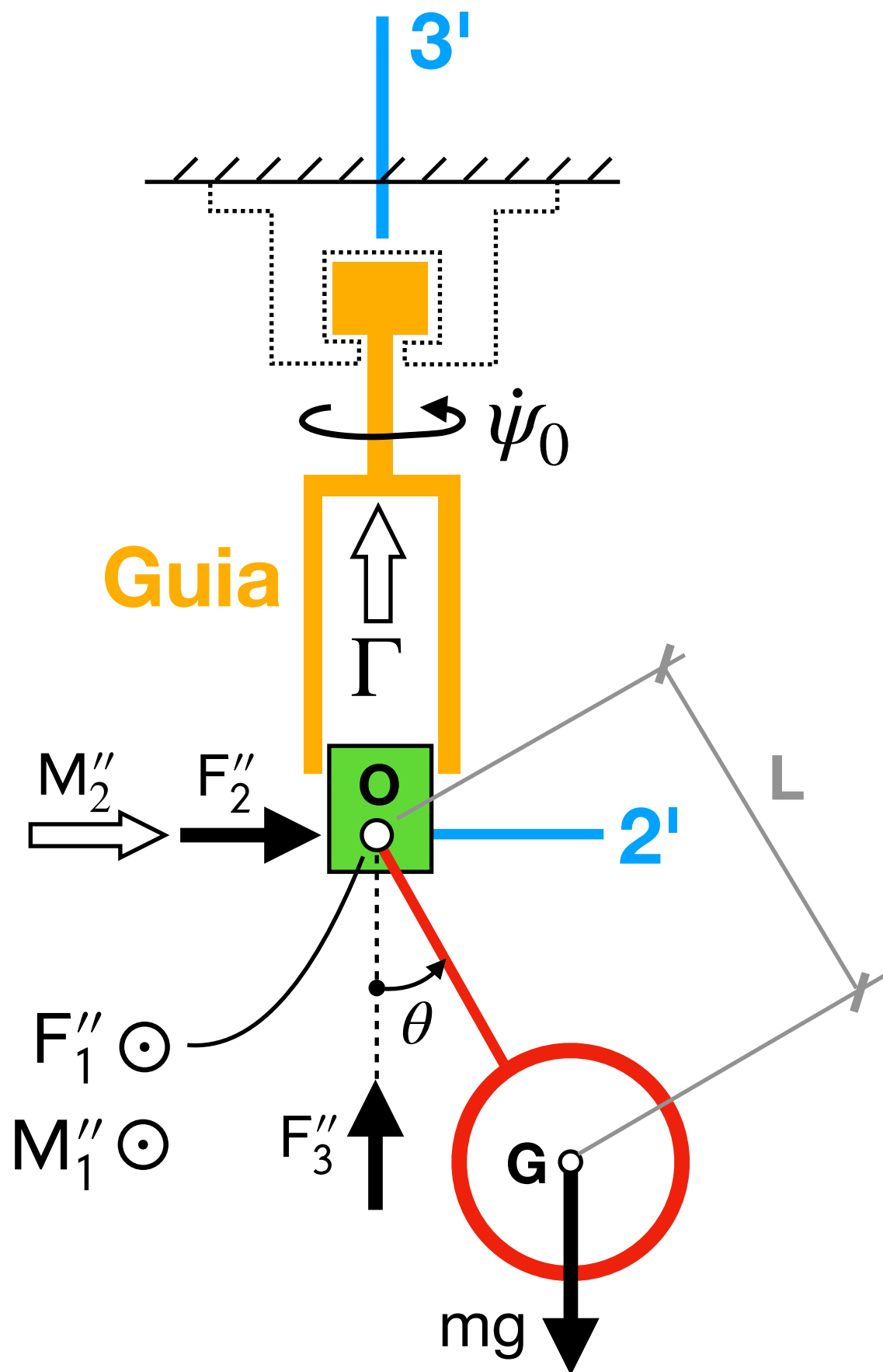


TMC(O)]₁ sobre SIST = Pèndol



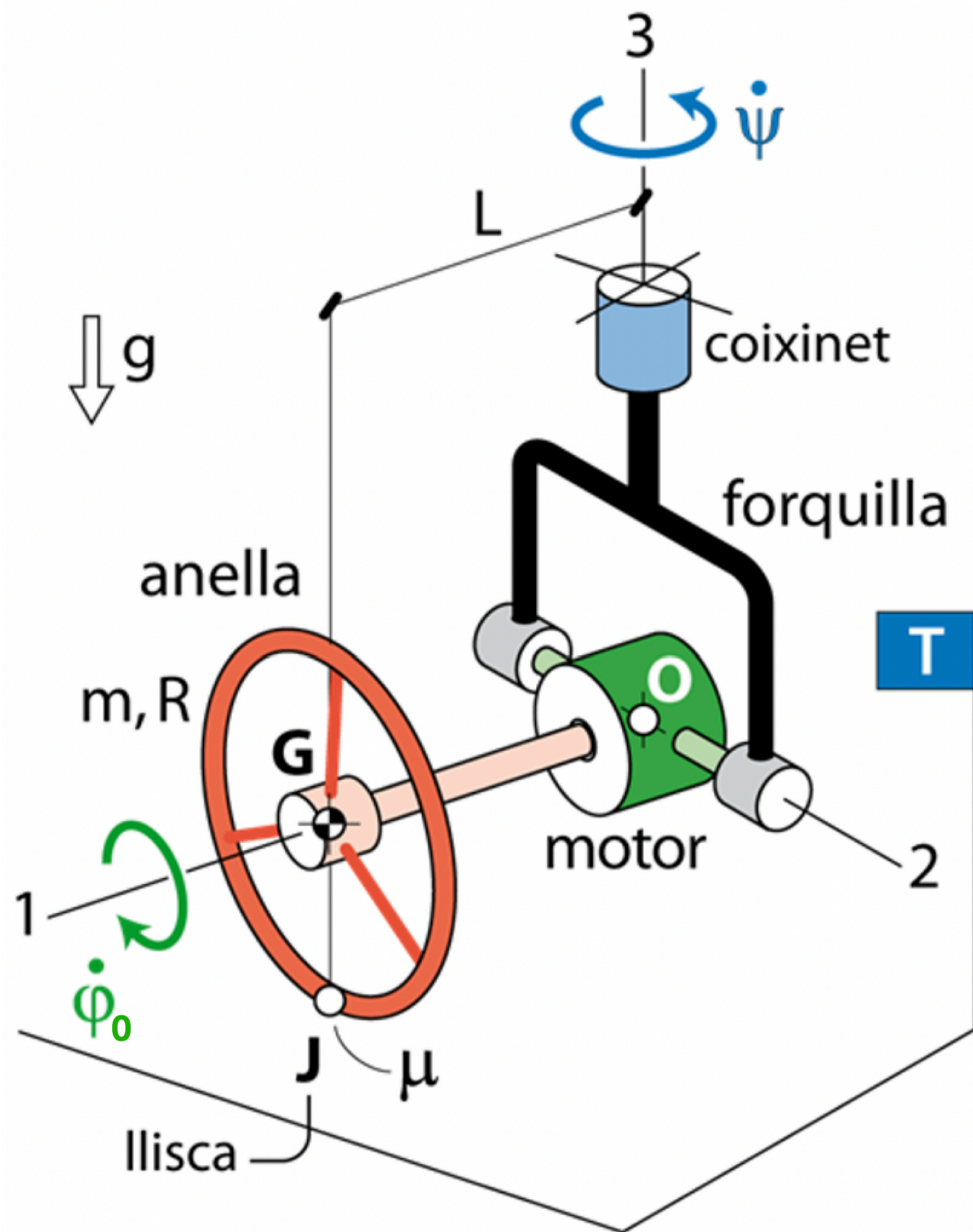
$$\left\{ \dot{\ddot{\mathbf{H}}}_{\text{RTO}}(\mathbf{O}) \right\}_{\text{B}} = \left\{ \begin{array}{l} I_{11} \ddot{\theta} + (I_{33} - I_{22}) \dot{\psi}_0^2 \sin \theta \cos \theta \\ (I_{11} + I_{22} - I_{33}) \dot{\psi}_0 \dot{\theta} \cos \theta \\ (I_{22} - I_{11} - I_{33}) \dot{\psi}_0 \dot{\theta} \sin \theta \end{array} \right\} \quad \text{(III)}$$

Parell motor per mantenir $\dot{\psi}_0 = ct$



D'abans

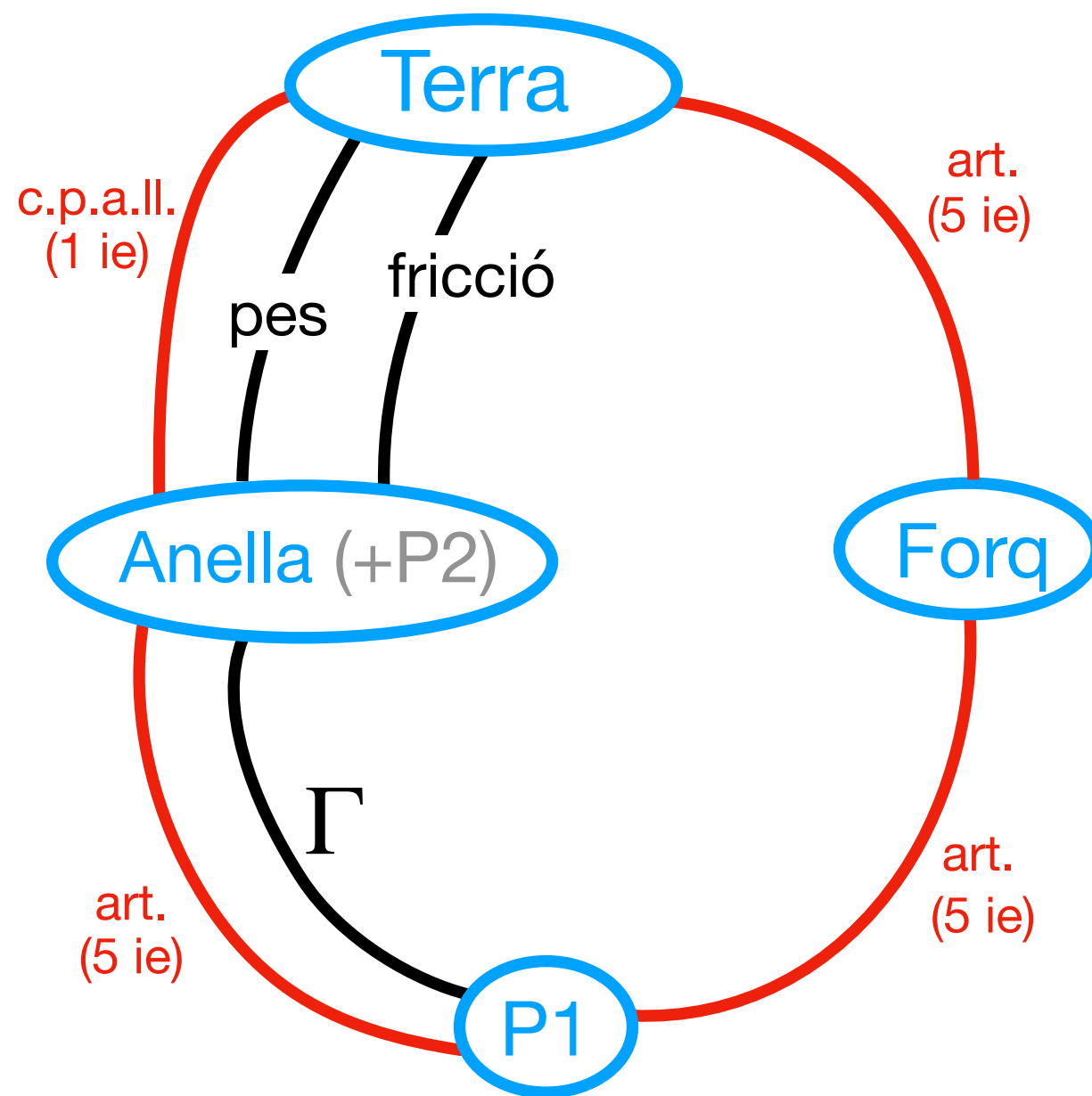
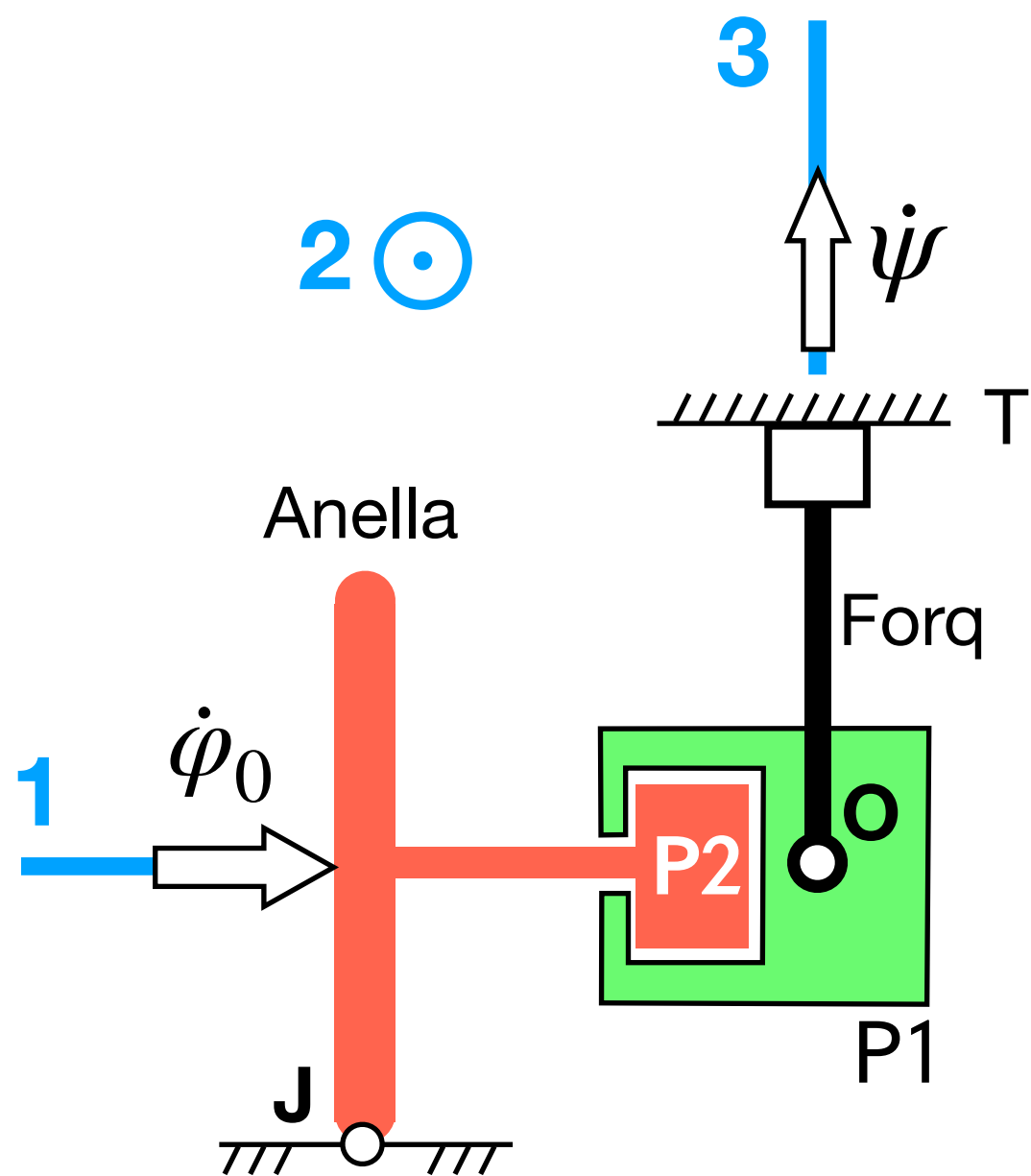
$$\left\{ \dot{\ddot{\mathbf{H}}}_{\text{RTO}}(\mathbf{O}) \right\}_{\text{B}} = \begin{Bmatrix} I_{11}\ddot{\theta} + (I_{33} - I_{22})\dot{\psi}_0^2 \sin\theta \cos\theta \\ (I_{11} + I_{22} - I_{33})\dot{\psi}_0 \dot{\theta} \cos\theta \\ (I_{22} - I_{11} - I_{33})\dot{\psi}_0 \dot{\theta} \sin\theta \end{Bmatrix} = \begin{Bmatrix} \dot{\mathbf{H}}_1 \\ \dot{\mathbf{H}}_2 \\ \dot{\mathbf{H}}_3 \end{Bmatrix}$$



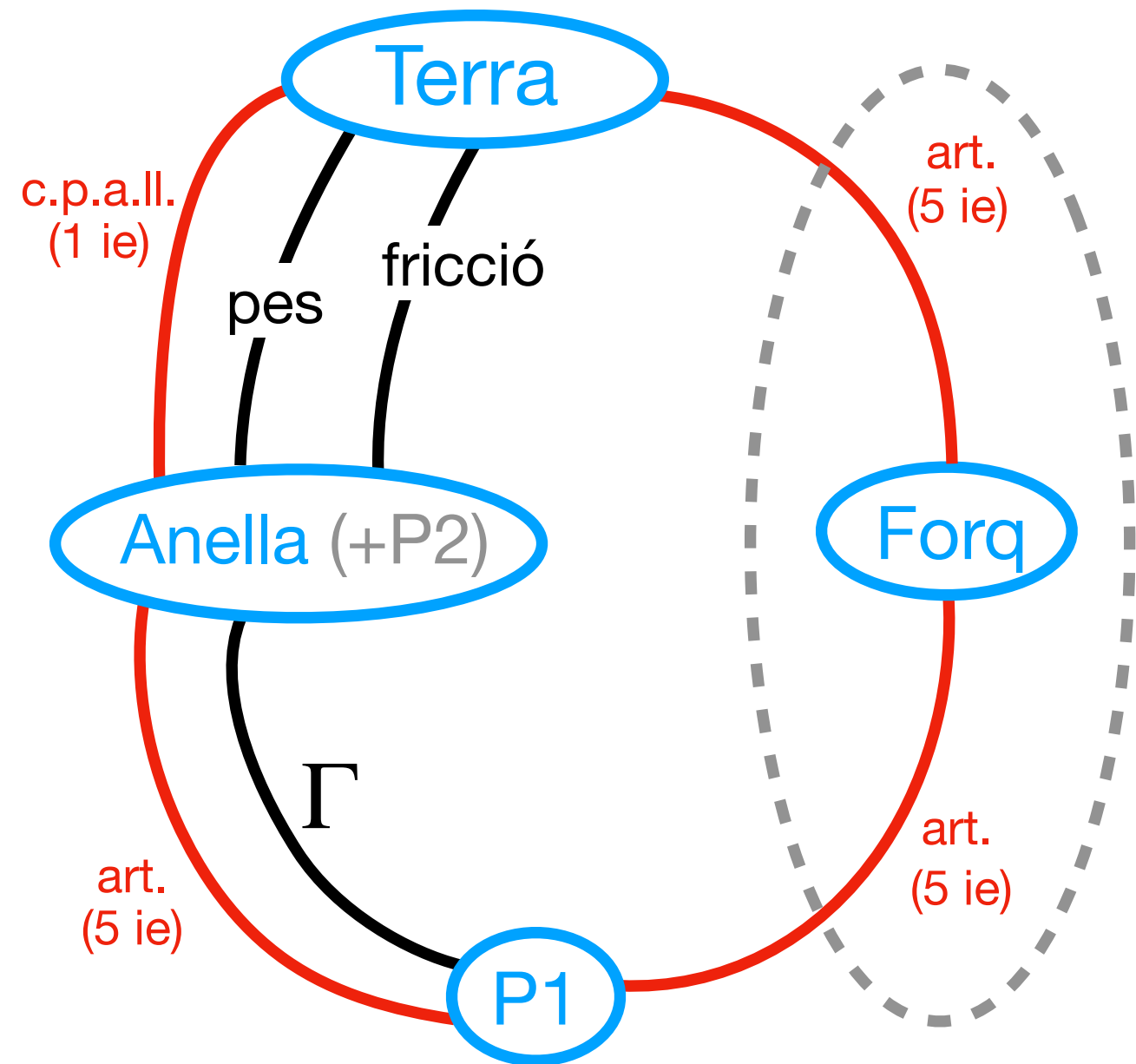
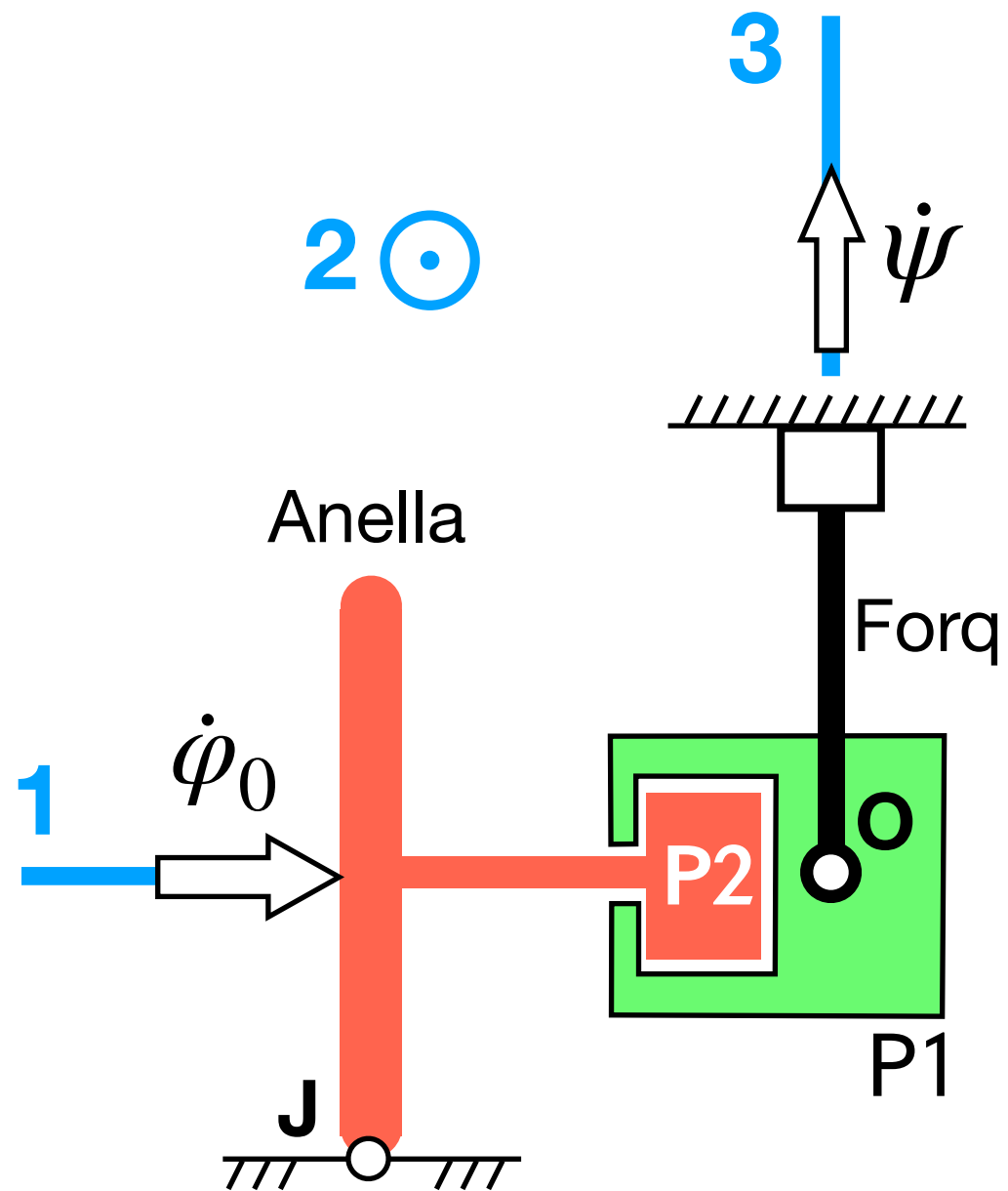
- GL sistema?
- DGI
- Caracterització de torsors?
- Eq. mov per a la coord. ψ
- Força normal a J
- Parell motor per mantenir $\dot{\phi}_0$

Manté contacte puntual a J

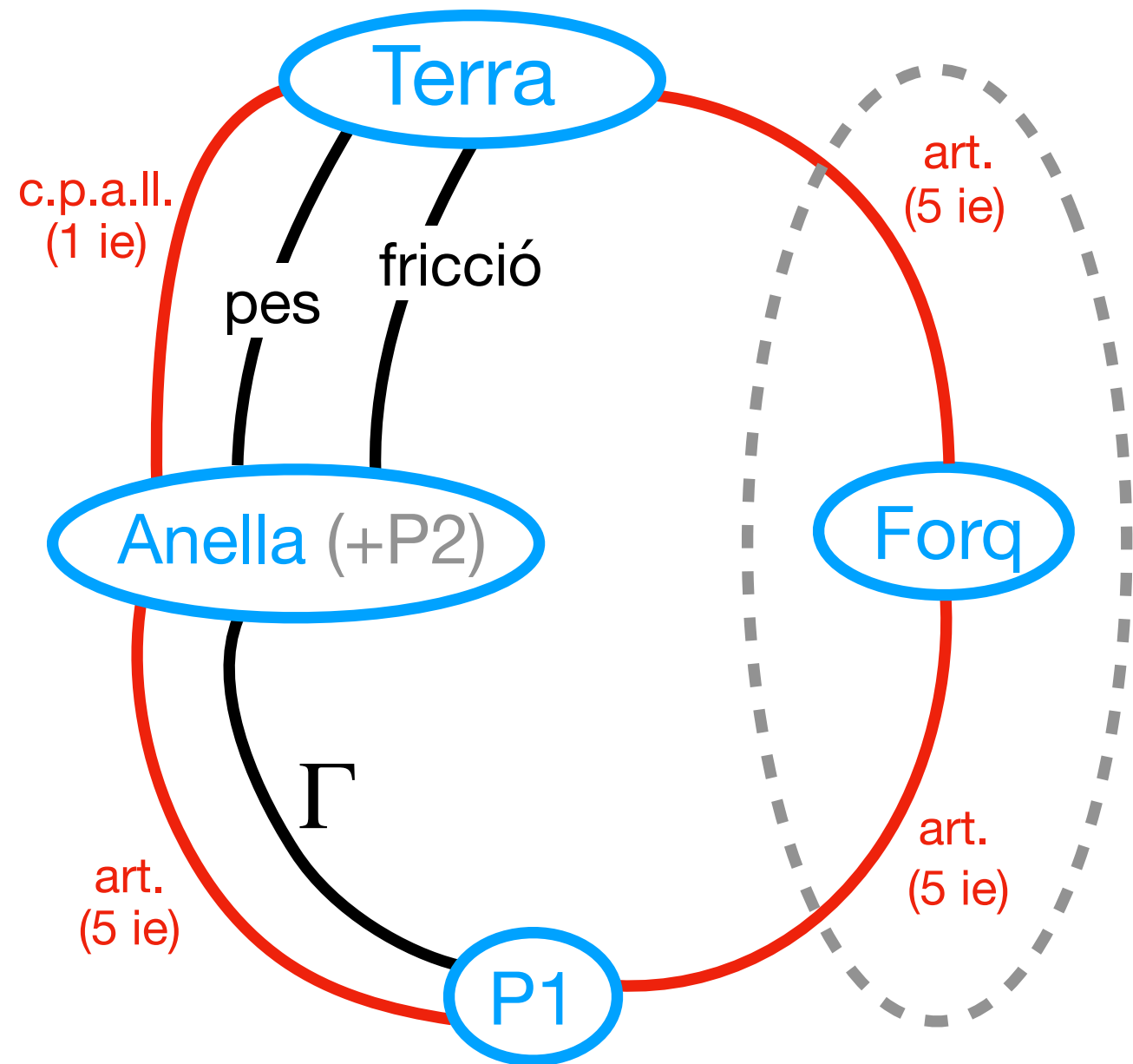
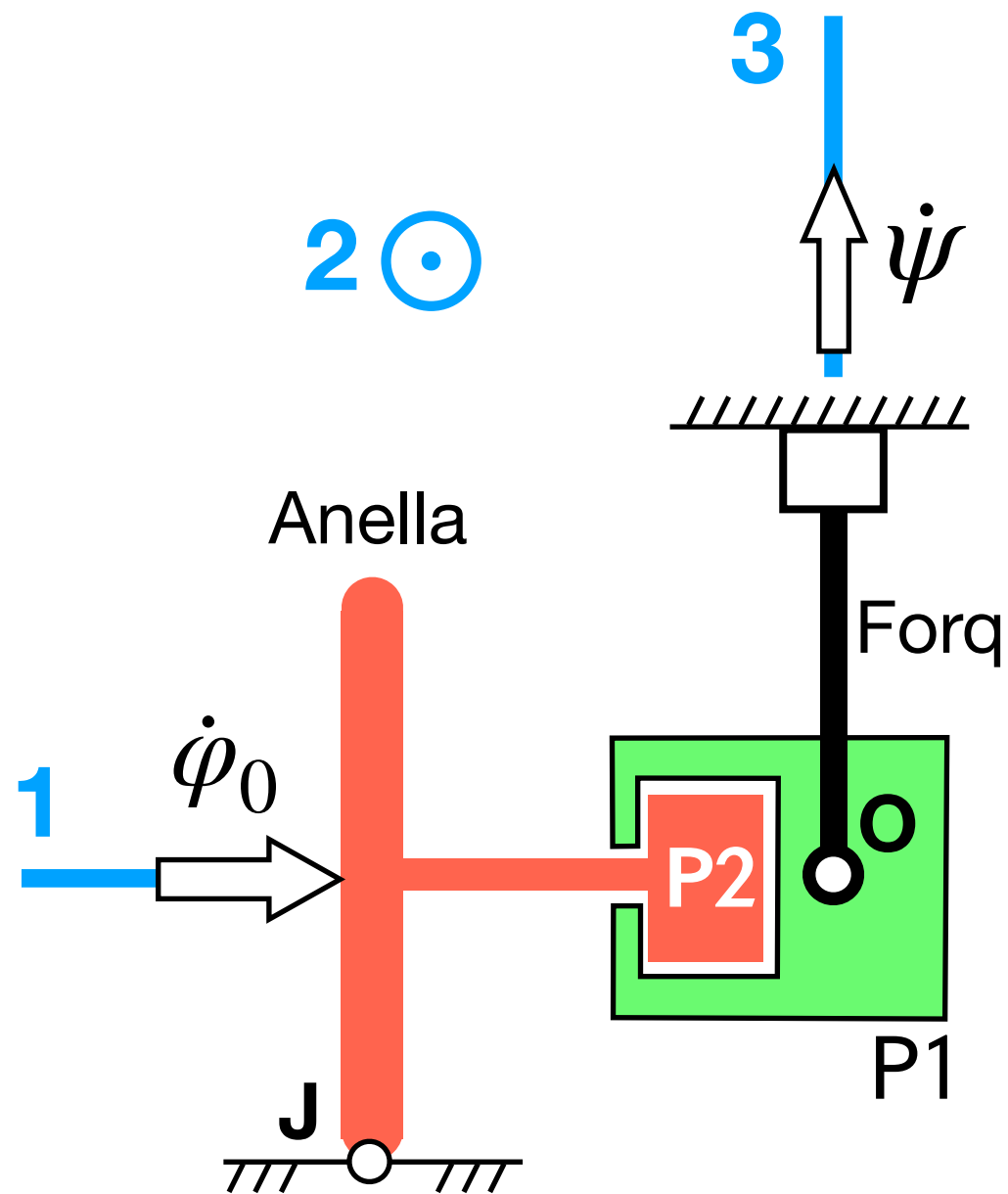
Motor manté $\dot{\phi}_0 = \text{ct}$



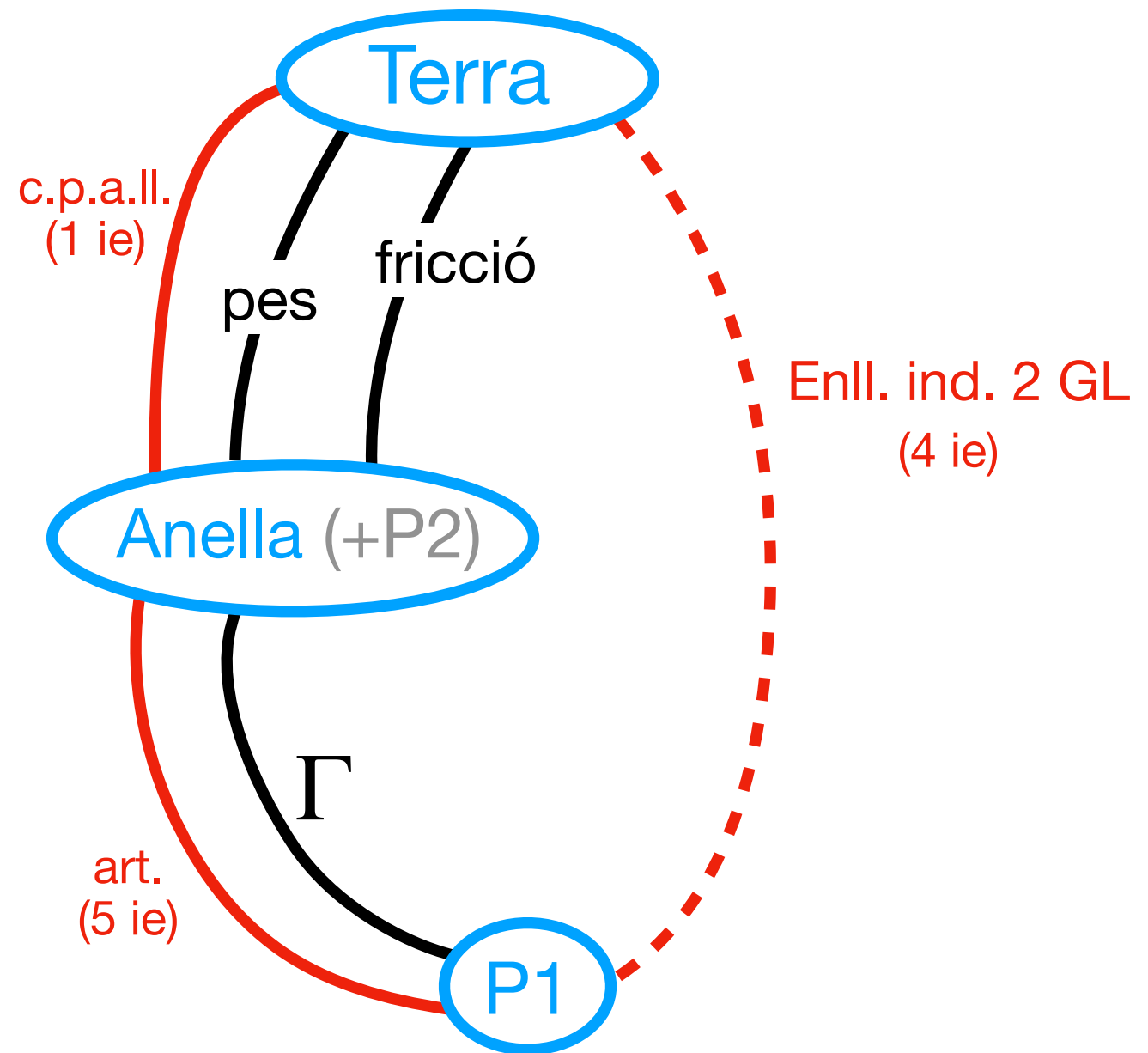
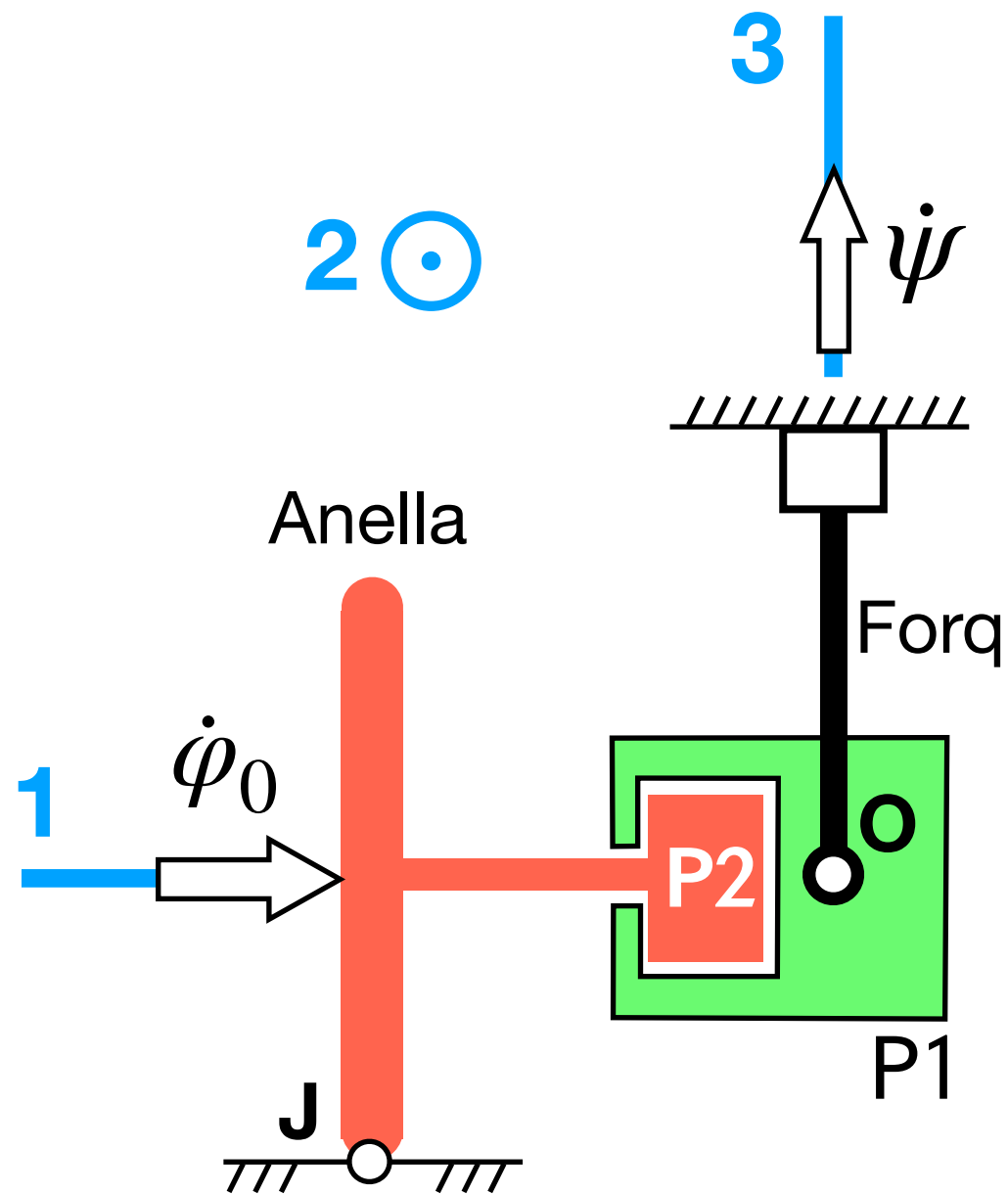
Forq és SAE ...



Substituim **Forq** per enllaç indirecte $T \rightarrow P_1$



Substituim **Forq** per enllaç indirecte $T \rightarrow P_1$



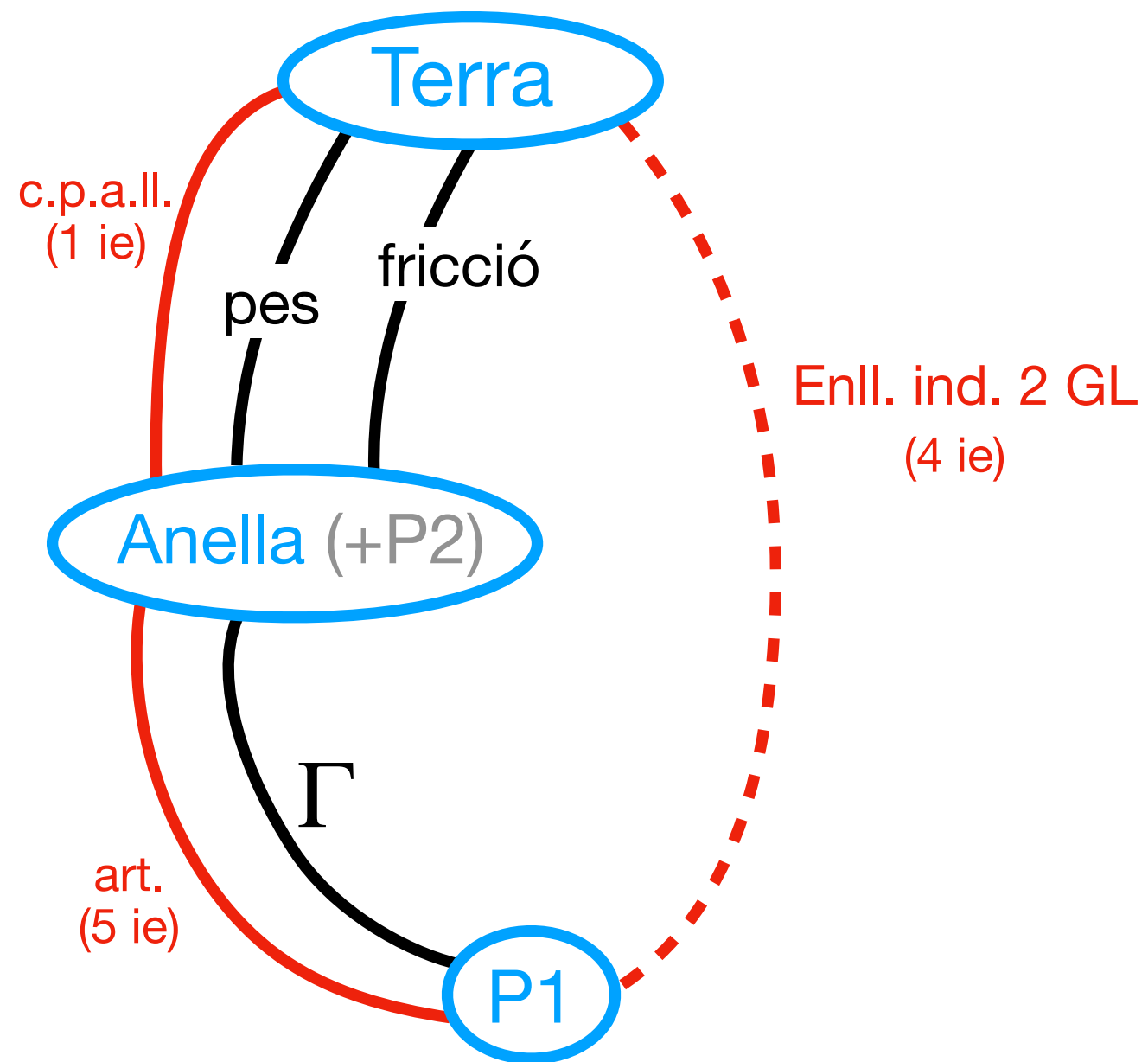
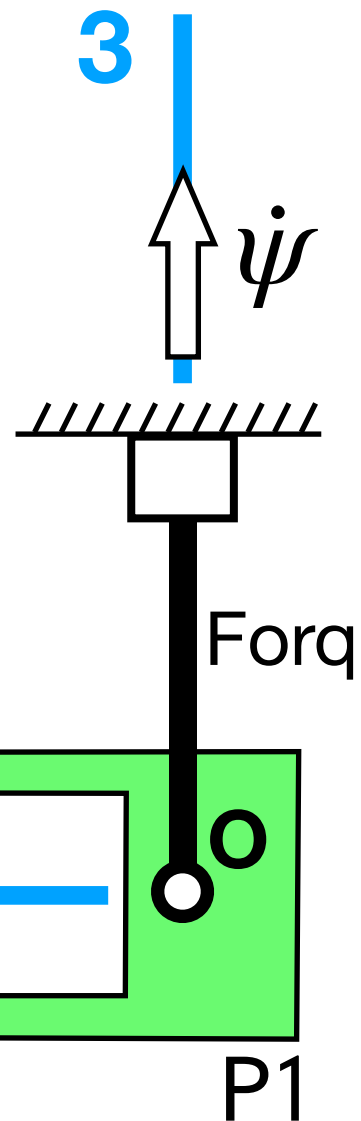
Substituïm **Forq** per enllaç indirecte $T \rightarrow P_1$

$B = (1, 2, 3)$

3

2 $\odot$

1

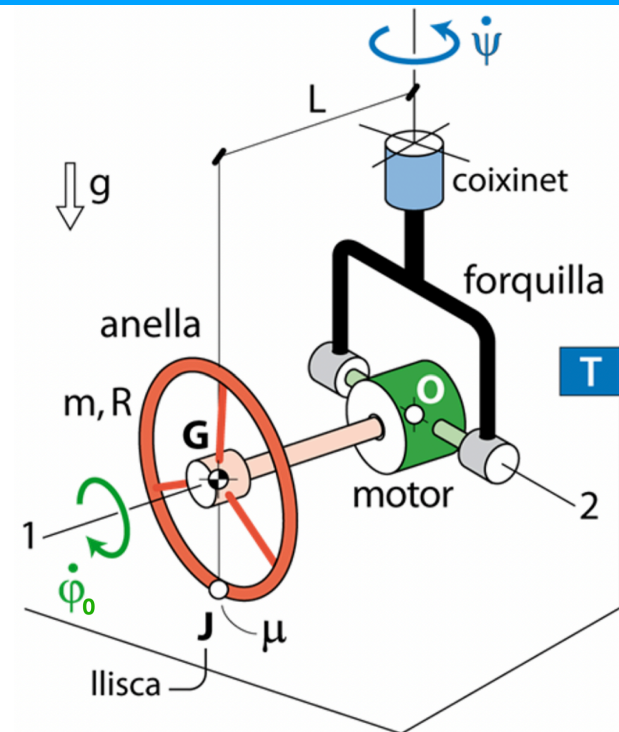


Torsor d'enllaç indirecte $T \rightarrow P_1$ al punt **O**:

$$\left\{ \bar{\mathbf{F}}_{T \rightarrow (\text{forq}) \rightarrow P_1} \right\}_B = \begin{Bmatrix} F_1 \\ F_2 \\ F_3 \end{Bmatrix} \quad \mathbf{O} \text{ fix a } T$$

$$\left\{ \bar{\mathbf{M}}_{T \rightarrow (\text{forq}) \rightarrow P_1} (\mathbf{O}) \right\}_B = \begin{Bmatrix} M_1 \\ 0 \\ 0 \end{Bmatrix} \quad \begin{array}{l} 0 \text{ en les rotacions} \\ \text{permeses de } P_1 \\ \text{resp } T \end{array}$$

Full ruta per eq. del mov. ψ

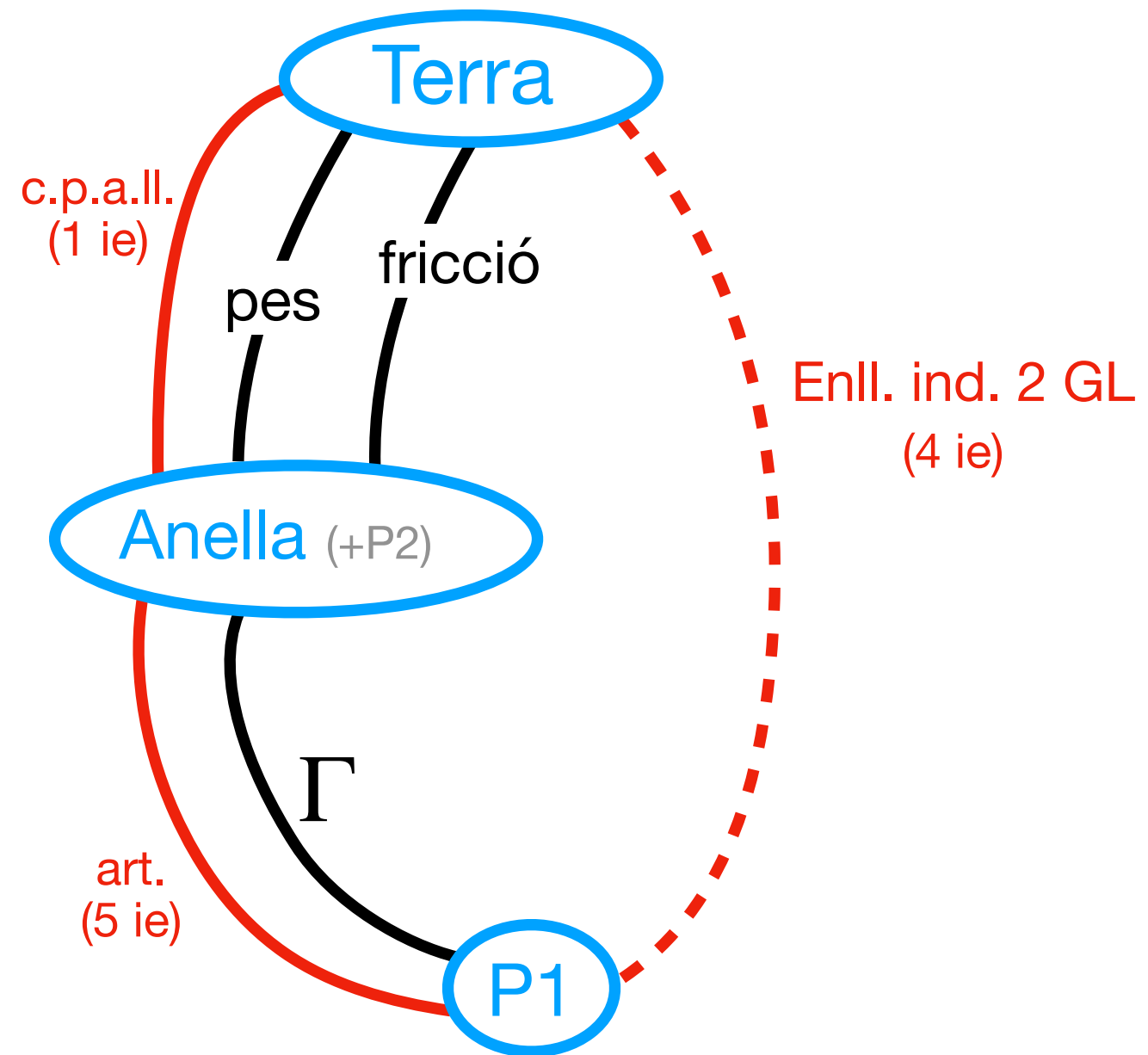


ψ afecta la cinemàtica de **P1** i **Anella**



Sist ha d'incloure **P1** o **Anella** (o ambdós)

Sistema	Incògn.	Problema
Anella	6 ie, Γ , $\ddot{\psi}$	INDET
P1	9 ie, Γ , $\ddot{\psi}$	INDET
Anella + P1	5 ie, $\ddot{\psi}$	DET



$\Rightarrow$ Triem SIST = Anella + P1